General arrangement of hydrodynamic power transmissions and their principle of operation (Fig. P-77)

The heart of a Föttinger™ converter is the hydraulic circuit that contains pump, turbine, and reaction member, all consolidated in a single casing and forming a closed fluid circuit.

The pump is connected to the input shaft, and the turbine to the output shaft. The fluid flow initiated by the pump drives the turbine. Power is transmitted by the circulation of the fluid between these two members of the converter, utilizing the mass forces of the circulating fluid.

Also with hydrodynamic power transmissions the sum of all torques must be zero. The reaction member absorbs the differential torque between input and output torques. Depending upon the torque acting on the guide blades, the turbine torque (output torque) may be larger or smaller than the pump torque or may be of the same magnitude (input torque). Under different operating conditions, the turbine speed may widely differ from the pump speed.

There is no mechanical connection between input and output ends. (See Fig. P-78.)

In gear units, the gears are correctly meshed and establish a force-locked connection between input and output ends (see Fig. P-79).

In hydrodynamic power transmissions, the circulating fluid connects input and output ends. No form-fit design, but a force-locked connection is used (see Fig. P-80).

Special features of hydrodynamic power transmissions

- Stepless transmission ratio (not constant)
- Flexible connection (no form-fit design)
- Load-controlled operation (the output speed matches the load on the output shaft)
- Transmission is free from wear and tear (no abrasion)
- Vibrational isolation (no mechanical connection between input and output ends)
- No reaction of output load on input end [by using suitable converter blading, free choice of driving motor (engine) with the required overload capacity; no stalling of engine or motor]

Figure P-81 shows a section of the tractive effort curve of a converter. The output speed is always adapted automatically to the prevailing load conditions. Figure P-82 shows the converter running driven equipment steplessly up to speed. Figure P-83 is meant to demonstrate that (since oil has no teeth) converters provide vibrational isolation.

Hydrodynamic power transmission operation

The ratio of input to output speed is not constant (as in the case of gear units) but adapts itself to the output load automatically and steplessly.

The absorbed power is determined by the characteristics of the torque converter.

The torques are not inversely proportional to the speeds as they are with mechanical transmissions.

Reversing the direction of rotation of pump and power flow provides a different behavior of the power transmission.

The converter types differ by the shape of their power absorption curves (absorbed power as a function of the ratio output speed/input speed).

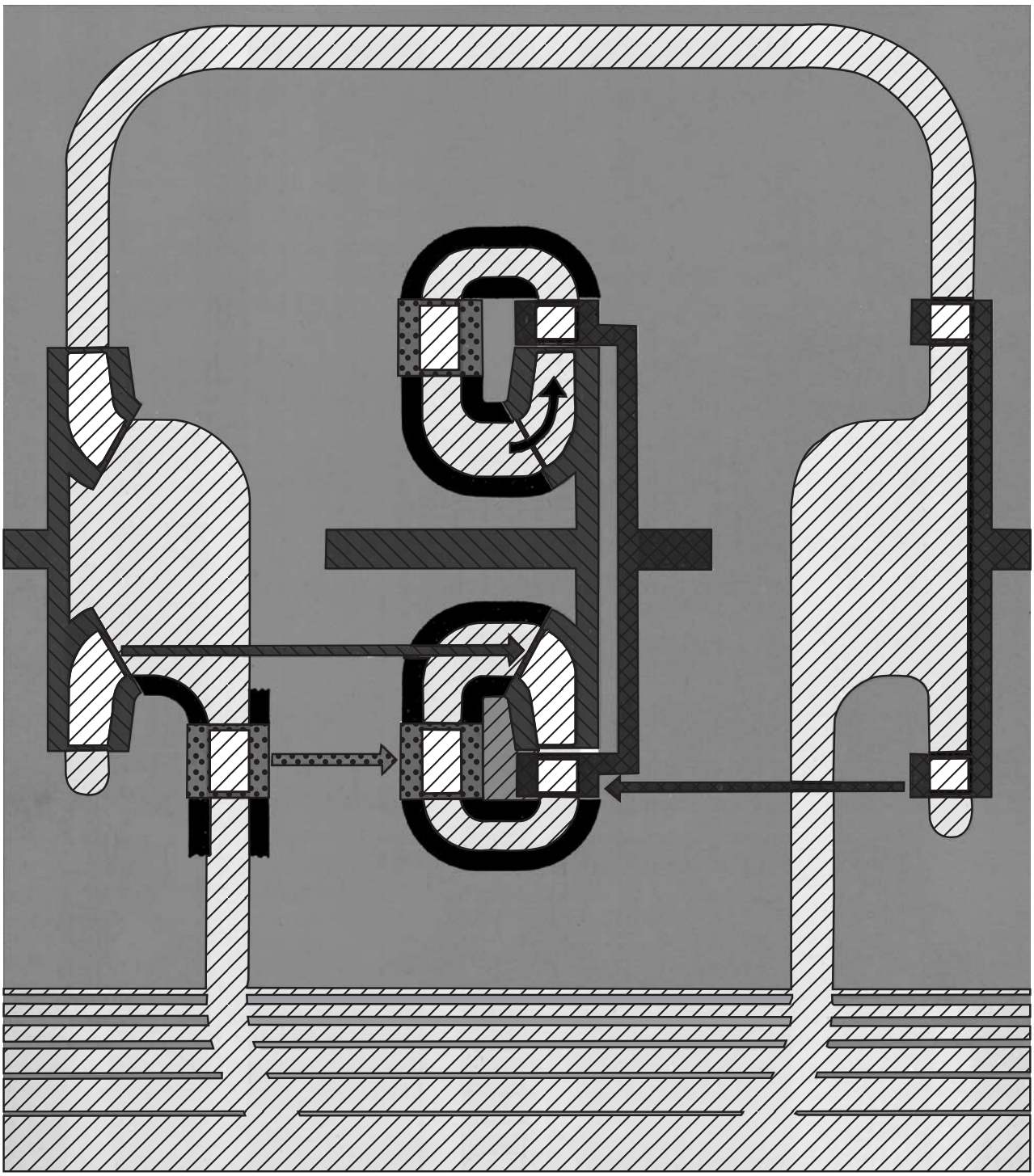


FIG. P-77 Hydrodynamic power transmission: operating principle schematic. (Source: J. M. Voith GmbH.)

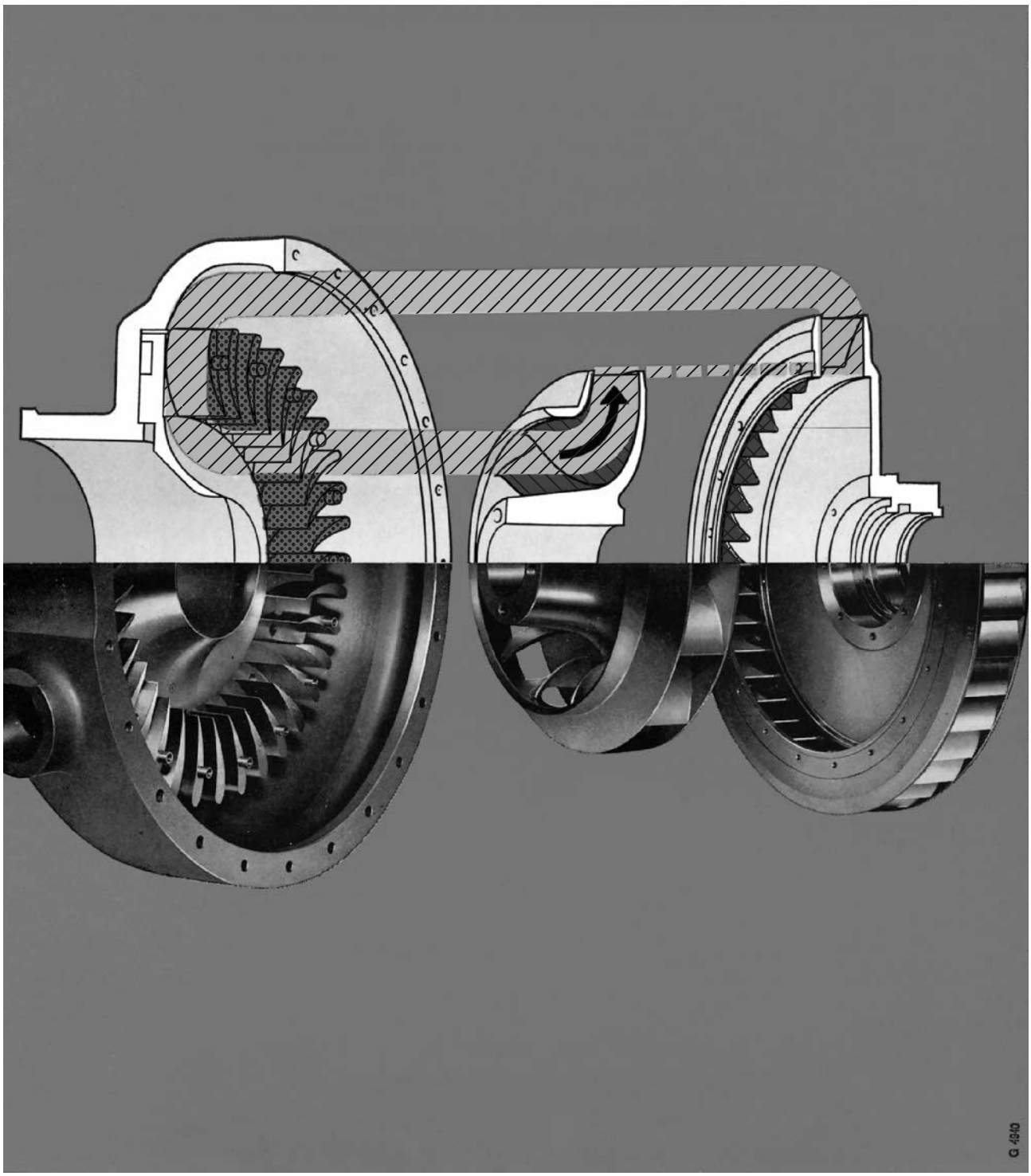


FIG. P-78 Hydrodynamic power transmission: operating principle cutaway. (Source: J. M. Voith GmbH.)

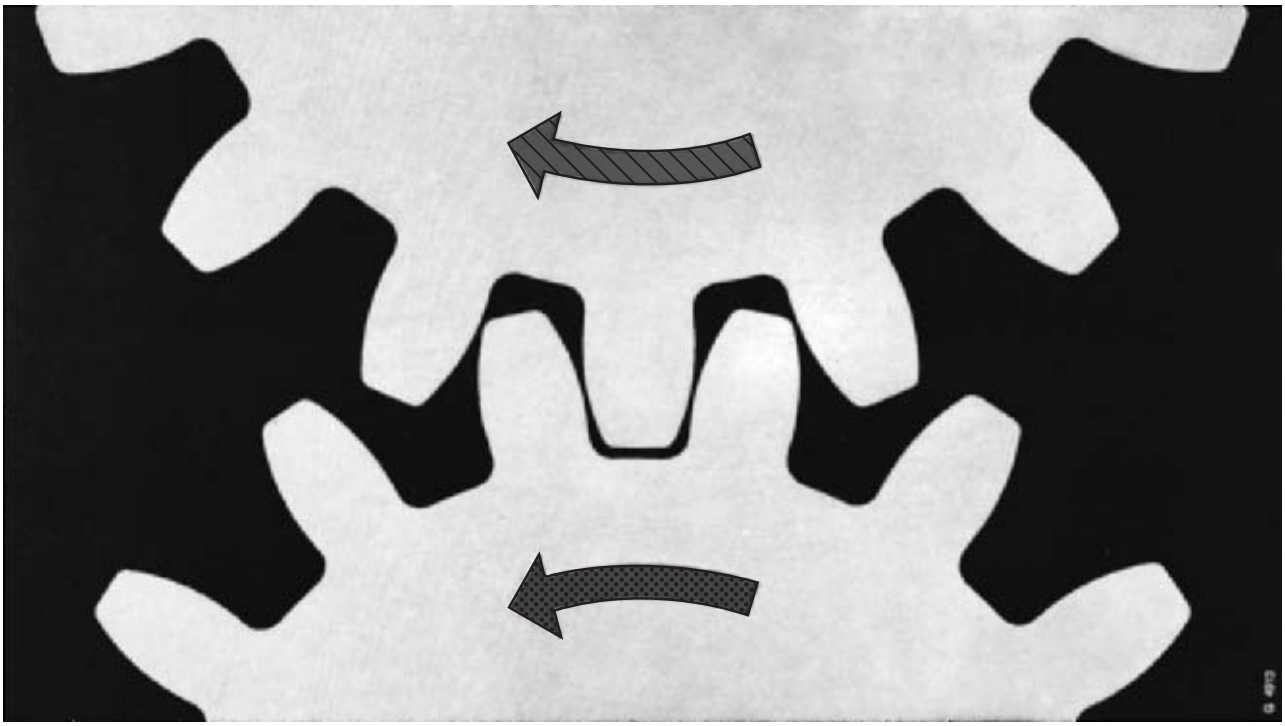


FIG. P-79 Gears: force-locked connection between gears. (Source: J. M. Voith GmbH.)

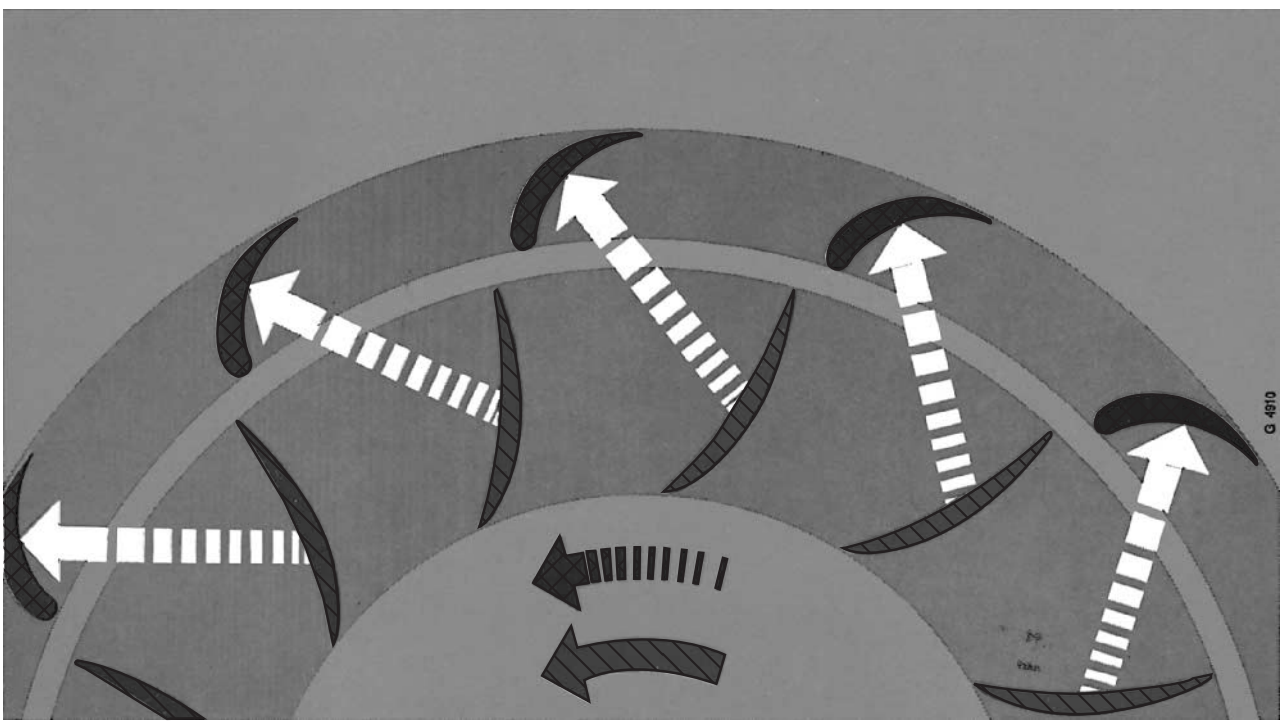
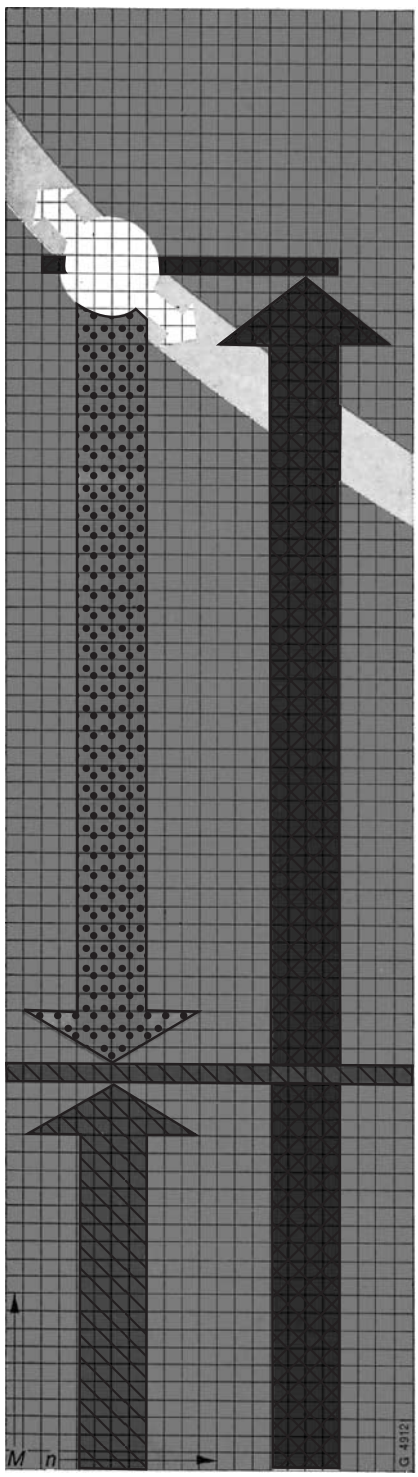
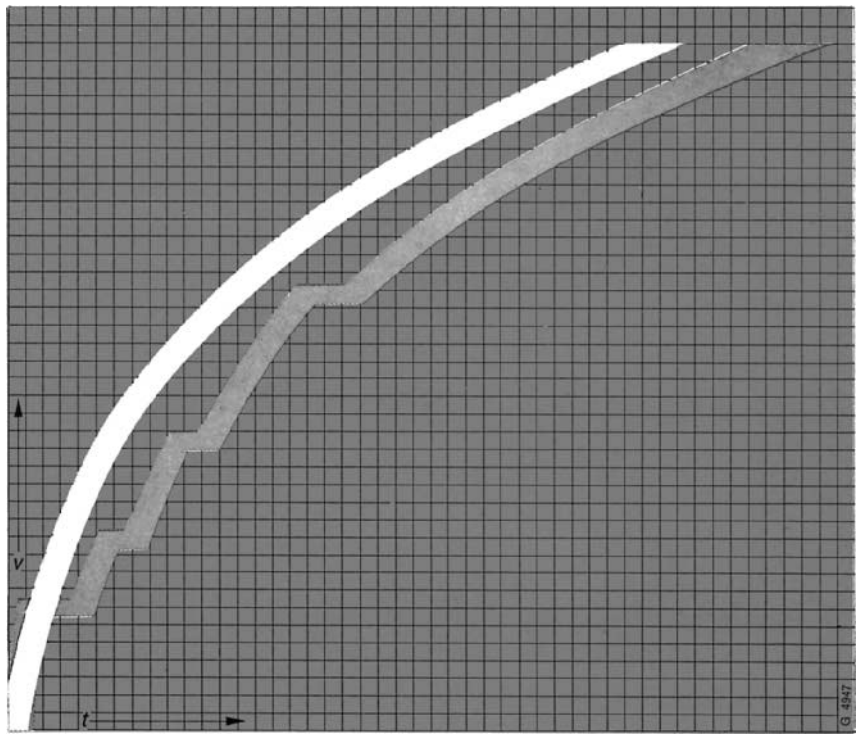


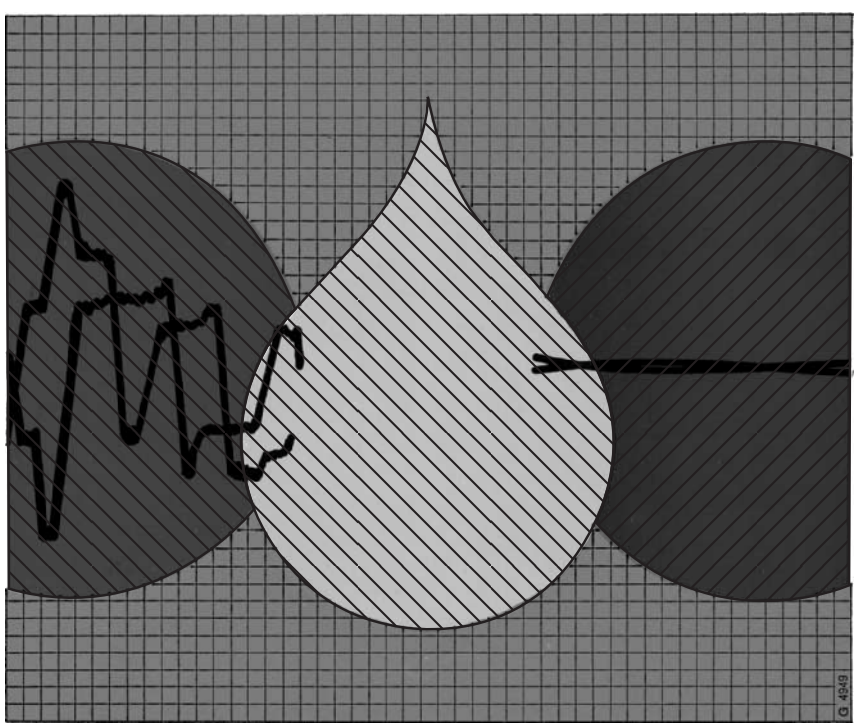
FIG. P-80 Hydrodynamic power transmission: circulating fluid provides connection between input and output ends. (Source: J. M. Voith GmbH.)



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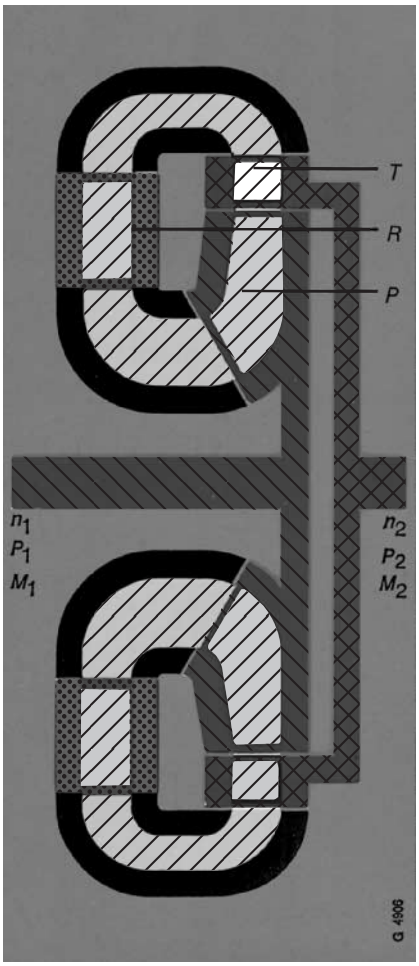


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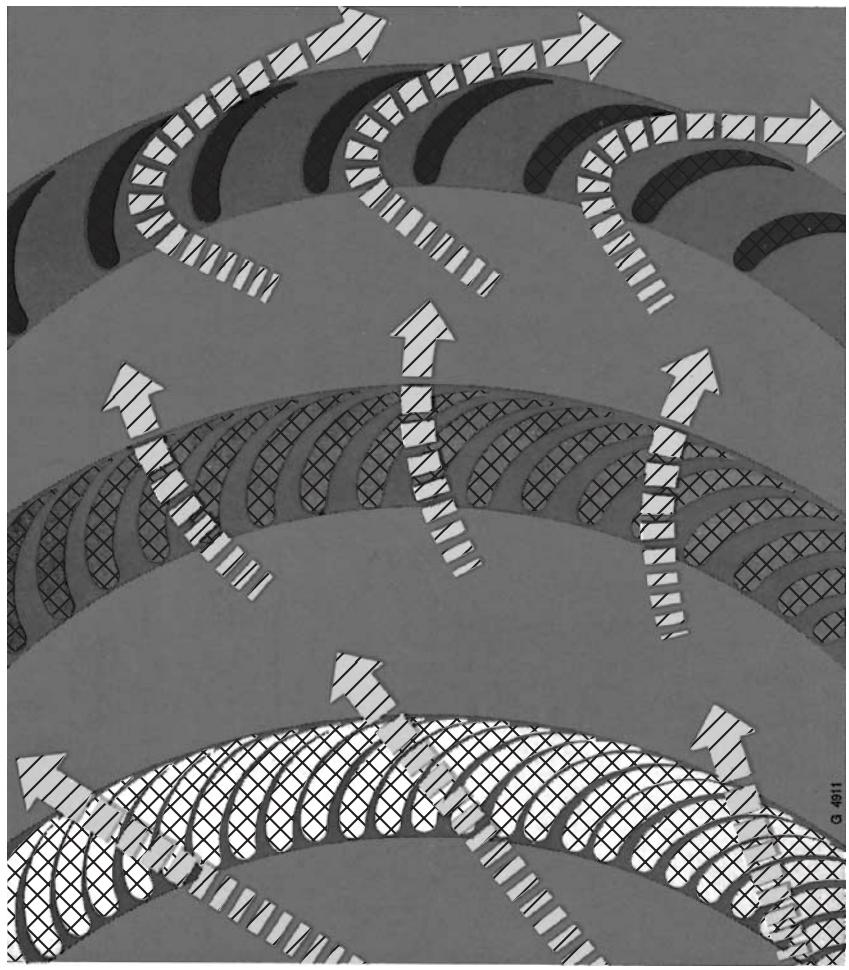
FIG. P-81 Part of the tractive effort curve of a converter. (Source: J. M. Voith GmbH.)

FIG. P-82 Hydrodynamic power can run driven equipment steplessly up to speed. (Source: J. M. Voith GmbH.)

FIG. P-83 As oil has no teeth, hydrodynamic converters provide vibrational isolation. (Source: J. M. Voith GmbH.)



84



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FIG. P-84 Schematic component of torque converter. (Source: J. M. Voith GmbH.)

FIG. P-85 Flow of operating fluid through turbine wheel under operating conditions. (Source: J. M. Voith GmbH.)

By acceleration (see Figs. P-84 and P-85) of a fluid mass inside the pump, a torque M_1 is created at the input shaft of the torque converter. The fluid mass is decelerated again in the turbine, thus developing a torque M_2 that is transmitted to the output shaft. Figure P-84 shows the schematic arrangement of the torque converter. Figure P-85 shows the flow of operating fluid through the turbine wheel under various operating conditions. Figure P-86 shows the converter's torque and efficiency curves (characteristics).

The stationary reaction member (guide blades) takes up the difference between input and output torque, thus providing torque multiplication. With the torque converter shown in the illustration, the absorbed torque M_1 is roughly constant with constant input speed n_1, even if the output speed n_2 fluctuates heavily. With increasing output speed, the torque M_2 at the output shaft steadily drops automatically and steplessly from a high startup torque. Any change in the

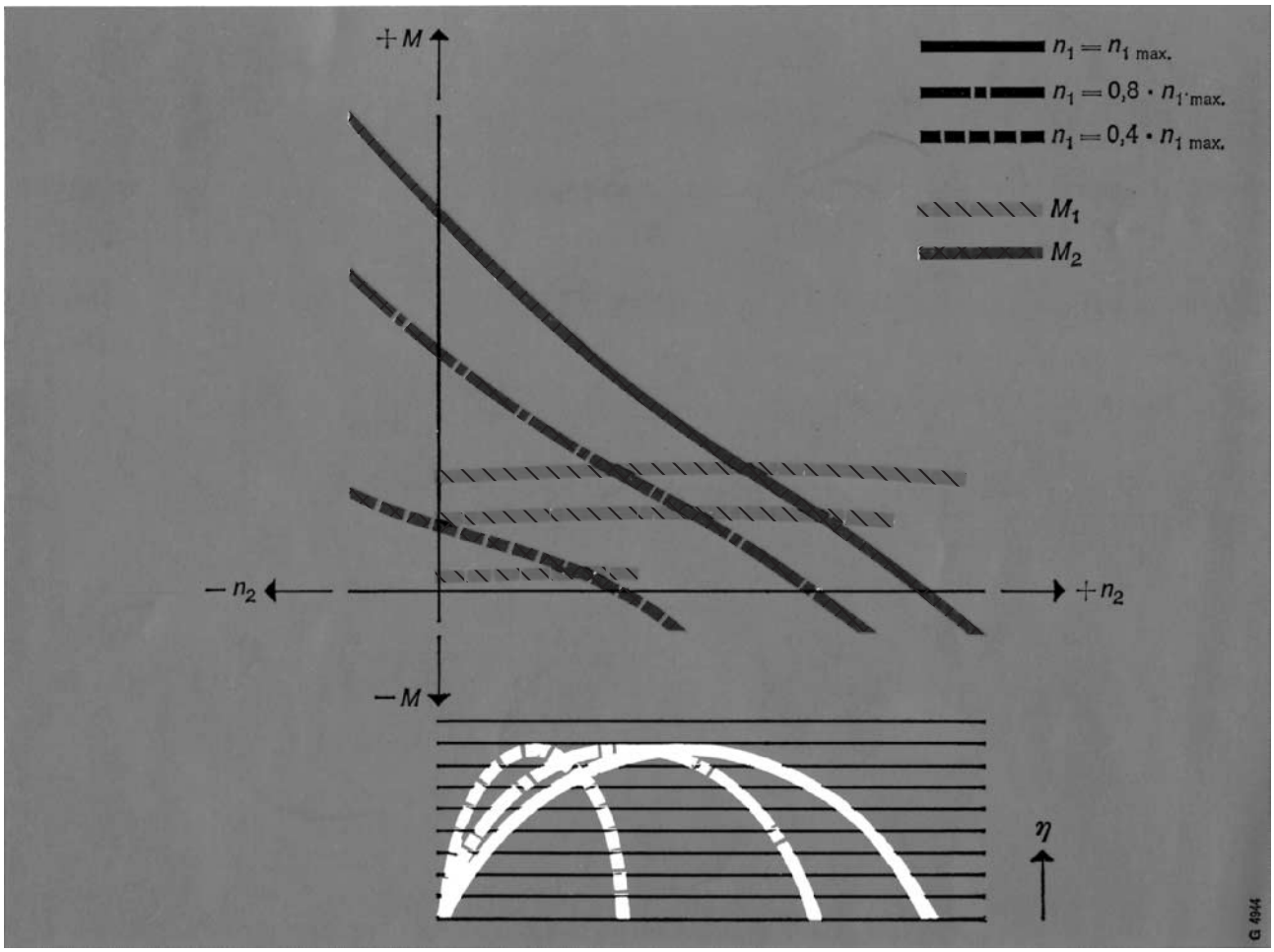


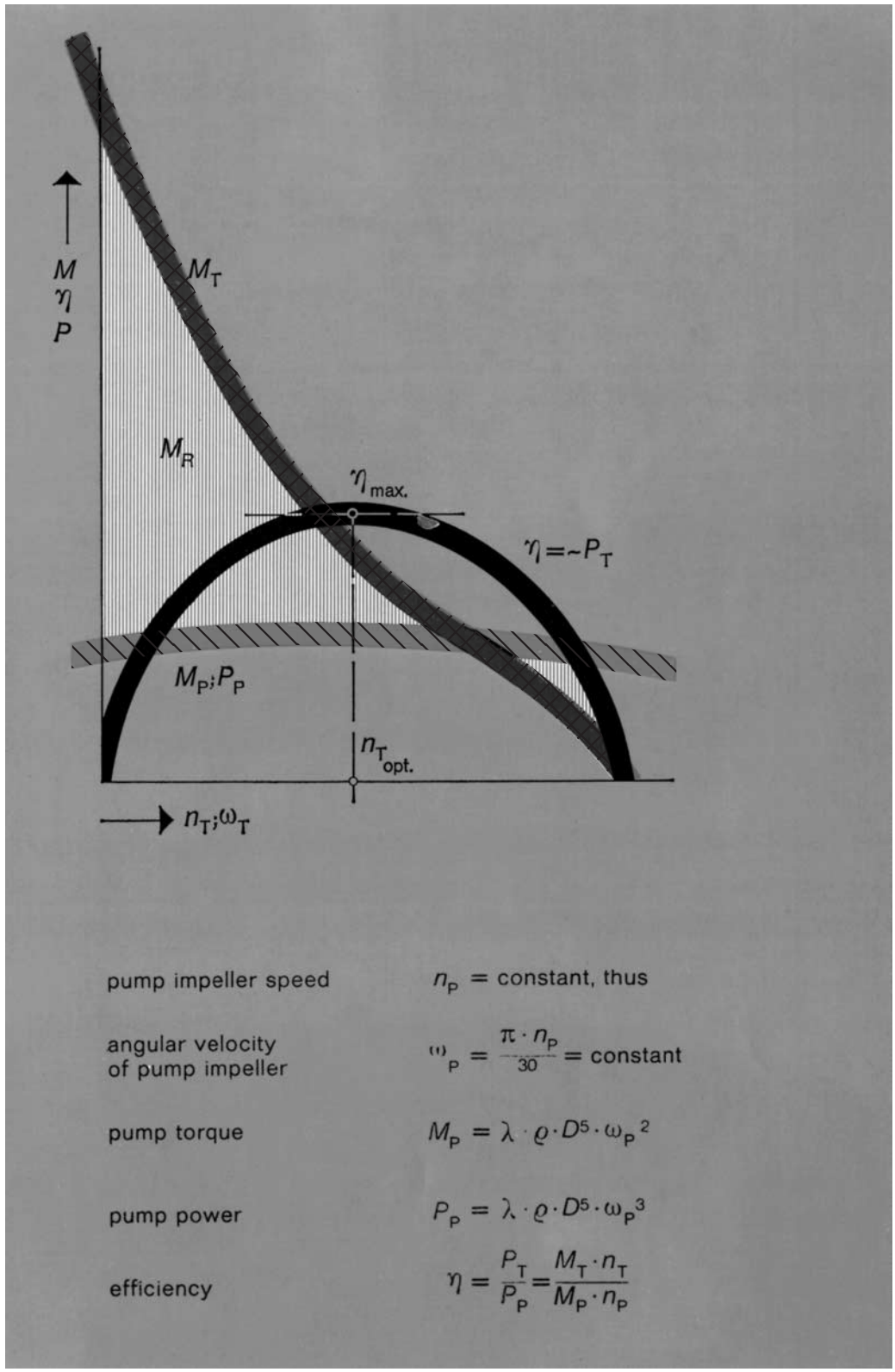
FIG. P-86 Torque and efficiency curve characteristics. (Source: J. M. Voith GmbH.)

deceleration of the fluid mass—due to a different turbine speed—also causes the transmitted torque to change. The circulating fluid is redirected by the turbine wheel, which causes the fluid to decelerate, and is shown for different operating conditions, viz. startup ($n_2 = 0$), rated speed ($n_2 = n_{\text{optimum}}$), and runaway speed ($n_2 = n_{\text{maximum}}$).

A change in output torque and output speed does not affect the motor (engine), even if the output speed should rise to such an extent that the output torque becomes zero or even negative. When the output speed is above the runaway speed, the torque converter produces a braking effect with no reaction on the motor (engine).

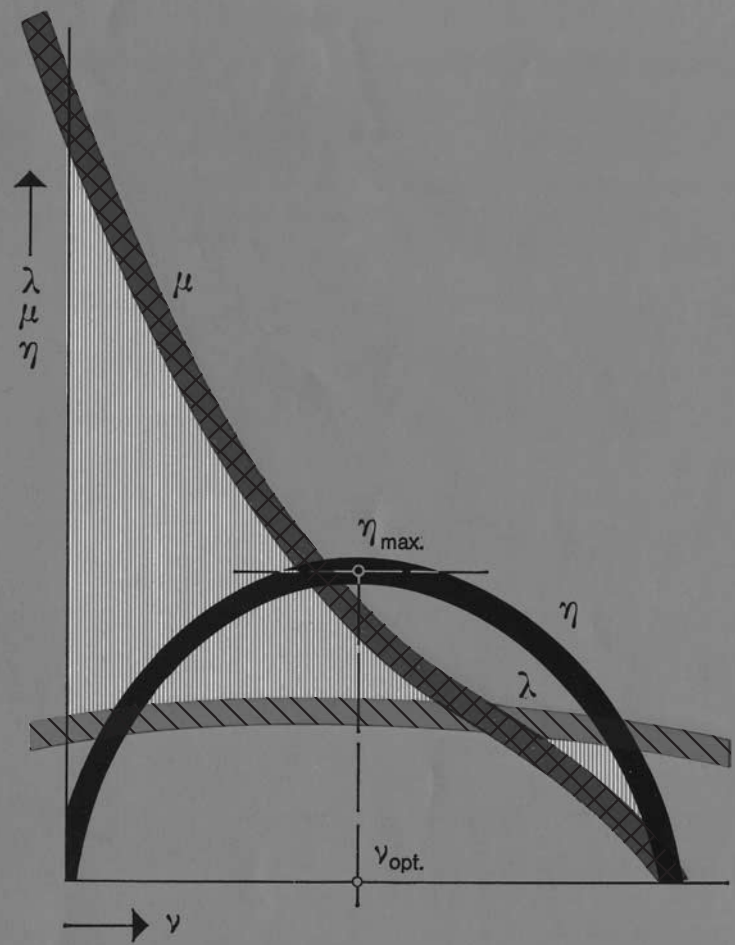
The characteristics of hydrodynamic power transmissions

See Figs. P-87 and P-88.



pump impeller speed	n_P = constant, thus
angular velocity of pump impeller	$\omega_P = \frac{\pi \cdot n_P}{30}$ = constant
pump torque	$M_P = \lambda \cdot \varrho \cdot D^5 \cdot \omega_P{}^2$
pump power	$P_P = \lambda \cdot \varrho \cdot D^5 \cdot \omega_P{}^3$
efficiency	$\eta = \frac{P_T}{P_P} = \frac{M_T \cdot n_T}{M_P \cdot n_P}$

FIG. P-87 Characteristics of a torque converter. (Source: J. M. Voith GmbH.)



speed ratio $\nu = \frac{n_T}{n_P} = \frac{\omega_T}{\omega_P}$

torque conversion $\mu = \frac{M_T}{M_P}$

efficiency $\eta = \frac{P_T}{P_P} = \frac{M_T}{M_P} \cdot \frac{\omega_T}{\omega_P} = \mu \cdot \nu$

power coefficient $\lambda = \frac{M_P}{\varrho \cdot D^5 \cdot \omega_P{}^2}$

specific gravity of operating fluid $\varrho = \frac{\gamma}{g}$

FIG. P-88 Dimensionless characteristics of a torque converter. (Source: J. M. Voith GmbH.)

Basic blading arrangements and associated converter characteristics

Power absorbed by converters is virtually constant. High torque multiplication possible. Suitable for motors (engines) that are sensitive to lugging down of their speed. See Figs. P-89, P-93, and P-97.

Main fields of application. Diesel locomotives and diesel railcars. Stationary drives with electric motors. Vehicles and construction machinery.

Power absorbed by converters drops at certain speeds. Clear limitation of maximum output speed. No overload protection required. See Figs. P-90, P-94, and P-98.

Main field of application. Road vehicles.

Power absorbed by converters drops. With increasing turbine speed, the power absorbed by the pump drops. The load on the driving motor (engine) increases with decreasing output speeds; the engine speed is lugged down. This results in fuel savings. See Figs. P-91, P-95, and P-99.

Main fields of application. Construction machinery. Shunting locomotives.

Power absorbed by converters increases. With increasing turbine speed, the absorbed power increases. Such characteristics are favorable for differential converters. See Figs. P-92, P-96, and P-100.

Main fields of application. Vehicles, in particular floor-level conveying equipment such as fork lift trucks, etc.

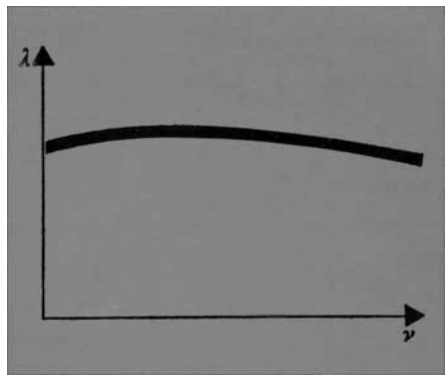
Operating costs comparison

Geared variable-speed turbocouplings reduce costs in conversions and new installations.

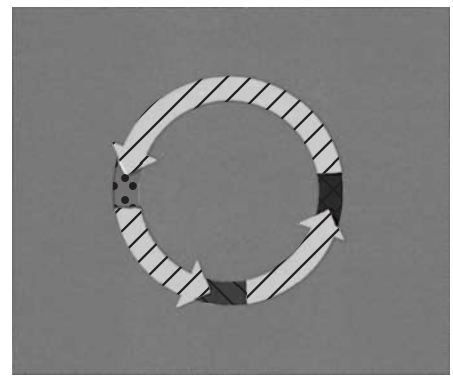
The generic advantages of a geared variable-speed turbocoupling are:

- A compact unit with integrated gear stage, designed and built to API613, SF1.4
- Motor starting under no load—stopping of the turbocompressor while the motor continues to run (rapid emptying)
- Controlled starting and run-up through critical speeds and process fields up to maximum compressor speed
- A wide infinitely variable-speed control range. Constant compressor output pressure in spite of varying molecular weight of the gas to be pumped
- Separate control of the starting and operating fields, each with control signals of 4–20 mA or 0.2–1 bar
- Energy saving compared with throttling on the suction side
- Damping of shock loads through hydrodynamic power transmission
- A simple unit requiring a minimum of maintenance and providing almost 100 percent availability
- Explosion-proof regulations can be inexpensively complied with
- The possibility of using standard squirrel cage motors

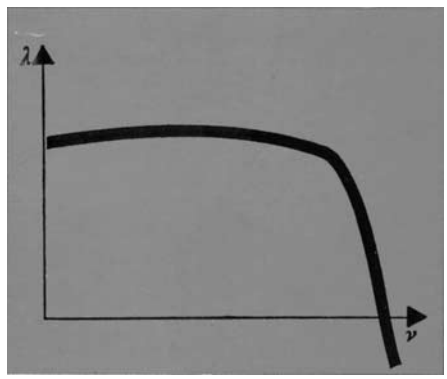
If optimal use is to be made of the advantages of hydrodynamic variable-speed couplings within an overall plant, then close cooperation is necessary between



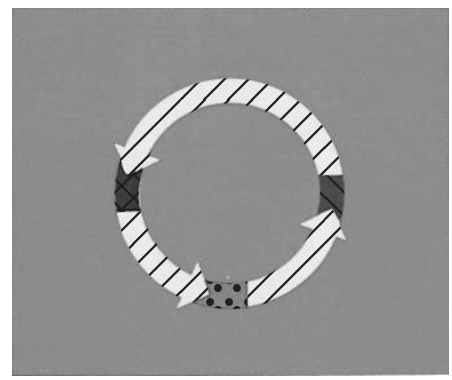
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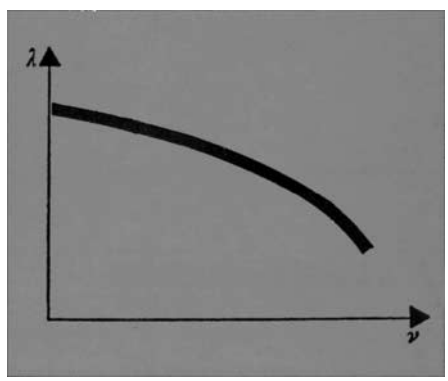
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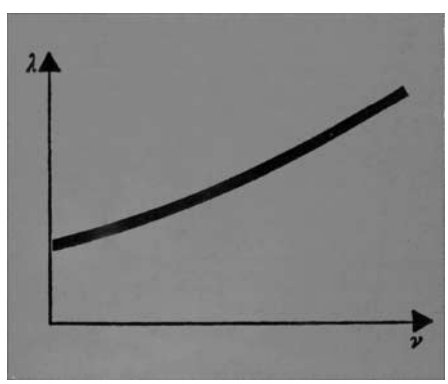
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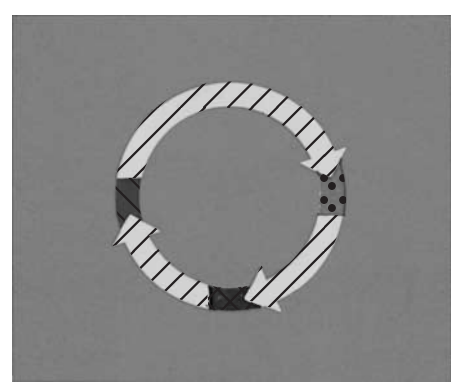
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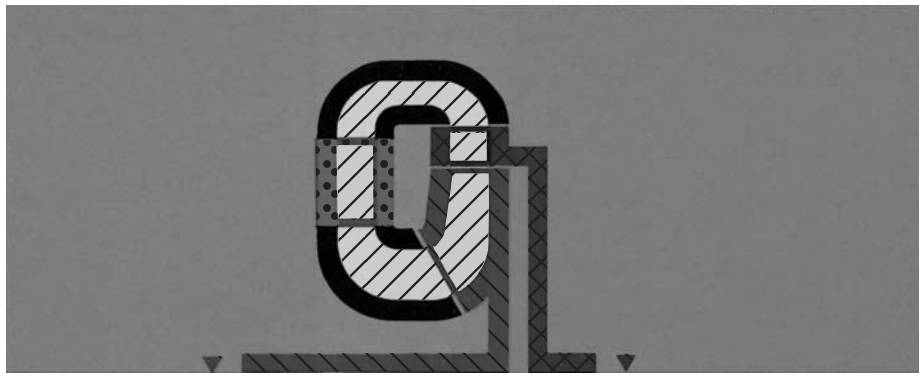
FIGS. P-89, P-93, P-97 Power absorbed by converters is constant. (Source: J. M. Voith GmbH.)

FIGS. P-90, P-94, P-98 Power absorbed by converters drops at certain speeds. (Source: J. M. Voith GmbH.)

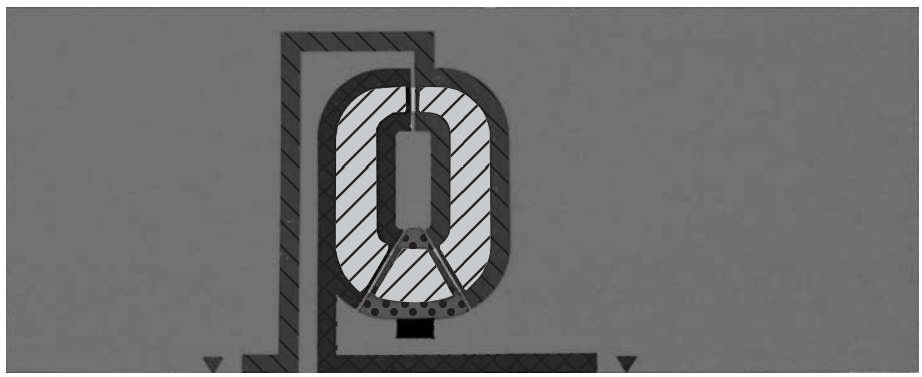
FIGS. P-91, P-95, P-99 Power absorbed by converters drops. (Source: J. M. Voith GmbH.)

FIGS. P-92, P-96, P-100 Power absorbed by converters increases. (Source: J. M. Voith GmbH.)

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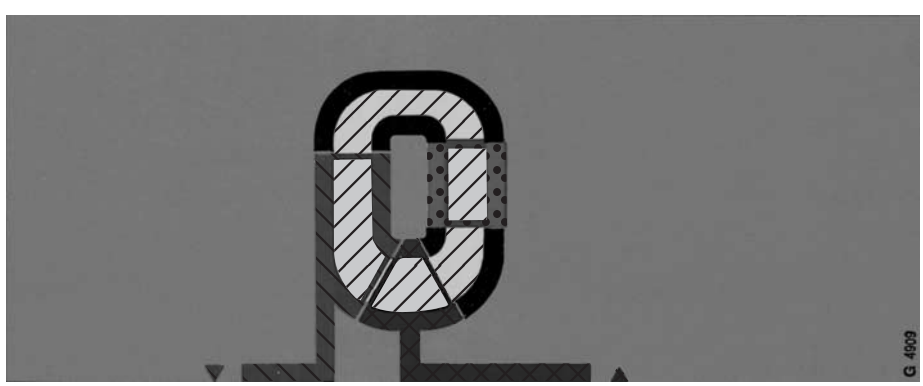
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the plant designer and the coupling manufacturer at the beginning of the project.

See also Figs. P-101 through P-103.

Scope for varying the characteristics and the connection of hydrodynamic power transmissions

The simplest version of a hydrodynamic power transmission has a constant operating-fluid filling. The converter characteristics are rigid and cannot be changed.

Influencing the converter circuit. (See Figs. P-104 through P-107.) Changing the characteristics of the converter requires suitable means in the hydraulic circuit. A slide valve may be used to throttle the flow of the circulating fluid. (See Fig. P-105.) With the slide valve closed, the startup of the motor or engine is facilitated. When the power flow is cut off, engagement of mechanical couplings will be eased.

Another possibility to change the characteristics is provided by adjustment of the guide blades, i.e., change of the position of the guide blades (reaction member). (See Fig. P-106.) Changing the guide-blade position varies the ratios of output/input speed and output/input torque.

Engaging and disengaging of converters can also be achieved by the filling and emptying principle. (See Fig. P-107.) Furthermore, by variation of the oil filling the transmitted power can be adjusted steplessly.

By applying this principle several converters can be used for opposite directions of running (hydrodynamic forward-reverse transmissions) and/or several speed ranges (hydrodynamic multicircuit transmission).

Influencing the circuit by using mechanical elements. (See Figs. P-108 through P-111.) The so-called "Trilok" converters represent a special type of converter: a freewheeling arrangement, which becomes effective at a well-defined speed ratio to prevent the guide blades from taking up a reaction torque, which converts the torque converter into a hydraulic coupling.

In some cases, it may be desirable to bridge the converter after the equipment has been run up to speed by the hydrodynamic converter. This can be accomplished by providing a direct mechanical drive.

If, for the prevailing operating conditions, the torque multiplication provided by the converter should not meet requirements, the operating range can be extended by a mechanical gearbox fitted behind the converter.

In the 2-channel arrangement of differential converters, the portion of the input power that is transmitted hydrodynamically decreases with increasing running speed while that transmitted mechanically increases until eventually the whole power is transmitted entirely mechanically. Differential converters are often followed by planetary gears.

Case studies*

Case study 1: Hydrodynamic variable-speed couplings in the petrochemical industry. In the course of recent years, use of hydrodynamic variable-speed couplings have proven to be an excellent means of controlling the speed of crude-oil and liquid-gas pumps in many production areas (including the Middle East and the North Sea). More than 150 geared variable-speed couplings and variable-speed couplings made by this

* Source: J.M. Voith GmbH, Germany.

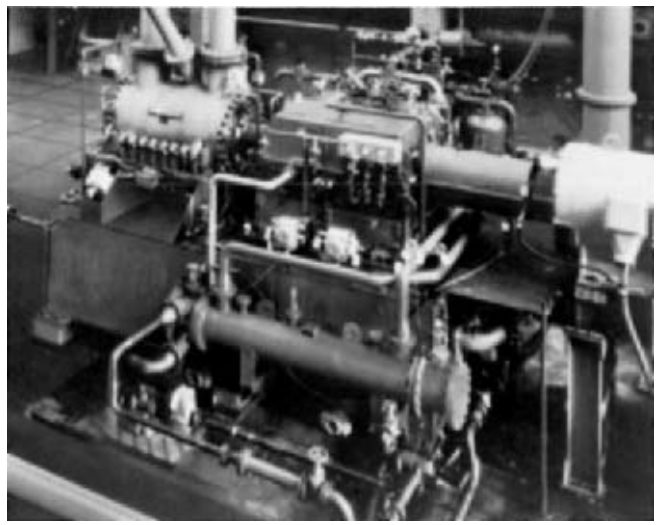


FIG. P-101 Compressor drive in a coal-pilot gasification plant. Type R18KGS—API, n_e = 1490 rpm, P = 7.3 MW, n_a = 10,548 rpm. (Source: J. M. Voith GmbH.)

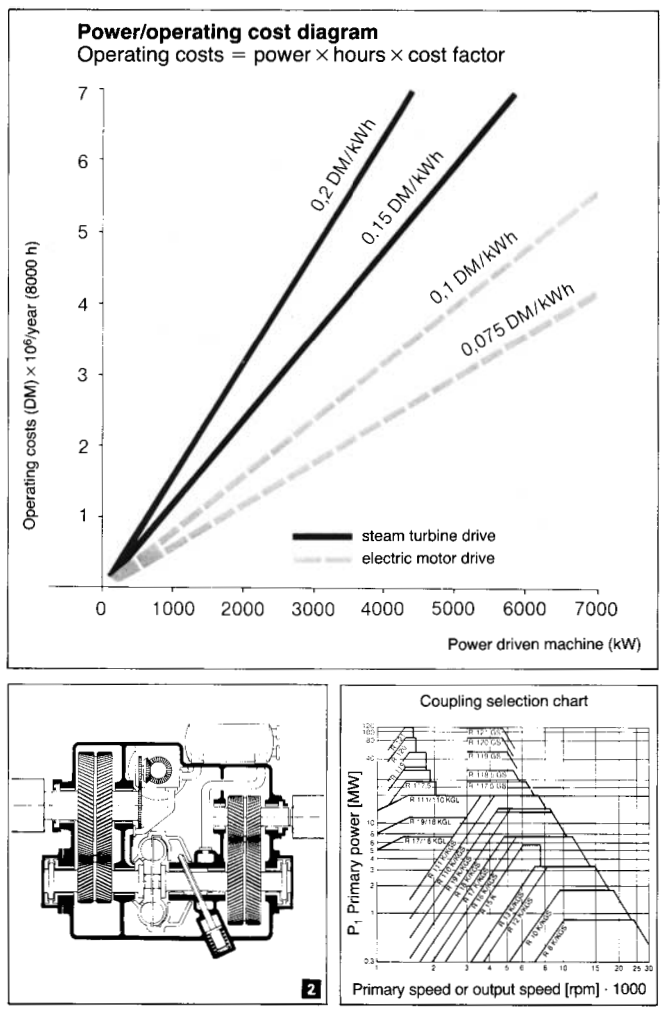


FIG. P-102 Design of a Voith hydrodynamic geared variable-speed coupling R1KGS (API) and coupling selection chart. (Source: J. M. Voith GmbH.)

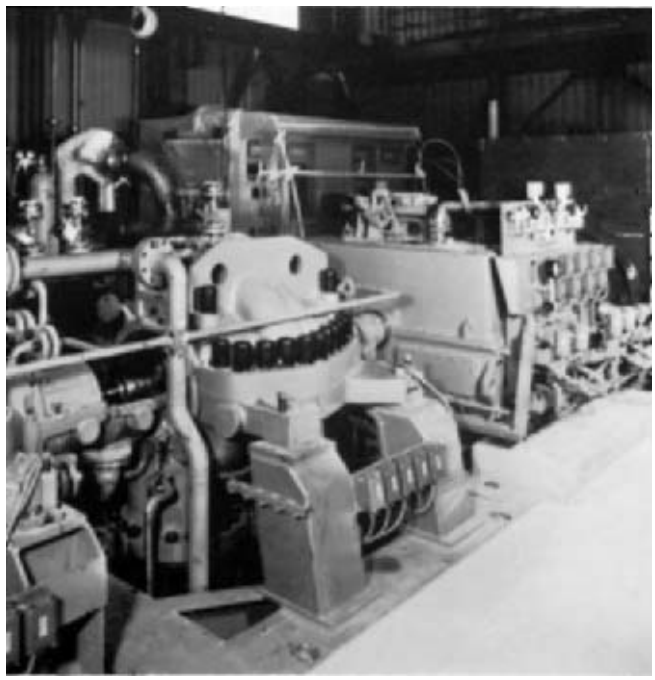


FIG. P-103 Compressor drive in an ethylene process plant. Type R16KGS—API, n_e = 1490 rpm, P = 2.9 MW, n_a = 12,604 rpm. (Source: J. M. Voith GmbH.)

information source are in use in crude-oil loading stations and pipeline pumping stations—often working under extremely severe environmental and operating conditions.

Operating periods of 15 years with as many as 50,000 hours per unit are not unusual. Transmission powers vary between 15 and 11,000 kW with pump speeds varying between 900 and 8000 rpm.

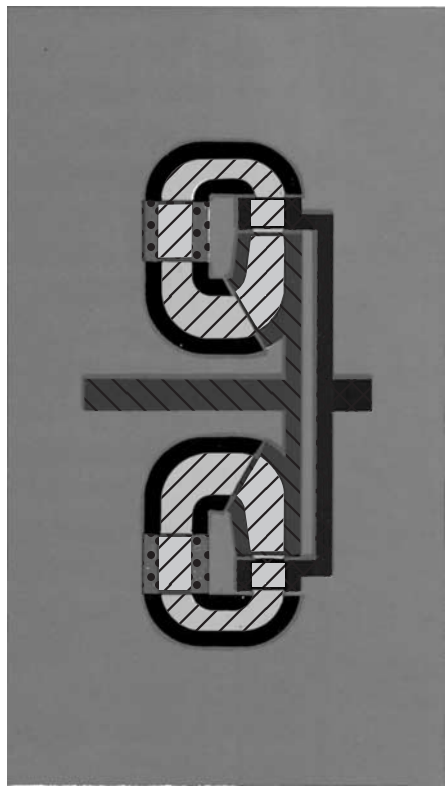
As previously noted, the variable-speed coupling made by this OEM is a hydrodynamic coupling (turbocoupling) that works on the Föettinger principle. The mechanical energy delivered by the drive motor is converted to kinetic energy by a pump impeller (primary wheel) and back to mechanical energy by a turbine wheel (secondary wheel) that is connected with the driven machine. Since there is no mechanical contact between the primary wheel and the secondary wheel, power is transmitted without wear. Mineral oils with suitable additives are used as operating fluid. See Fig. P-112.

The quantity of fluid in the variable-speed coupling can be varied at will during operation by means of a scoop tube and thus it is possible to adjust the power output level to meet the requirement at any time. The speed of the driven machine can be infinitely varied.

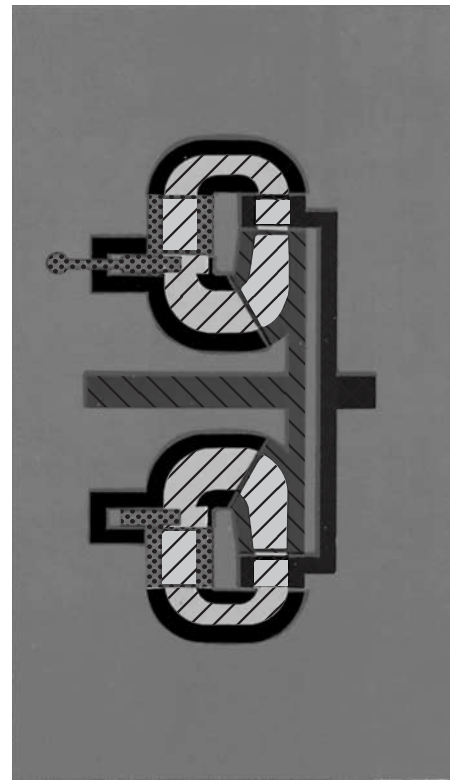
Various arrangements—with or without a mechanical gear stage—provide for adapting the coupling to the required speed.

Type SVNL. Type SVNL variable-speed couplings can be supplied either with plain or roller bearings. A particular design feature is the division of the coupling housing by a horizontal flange. This is necessary where the internal parts are large in size and it is often called for on smaller couplings in order to accommodate special installation conditions at the location concerned.

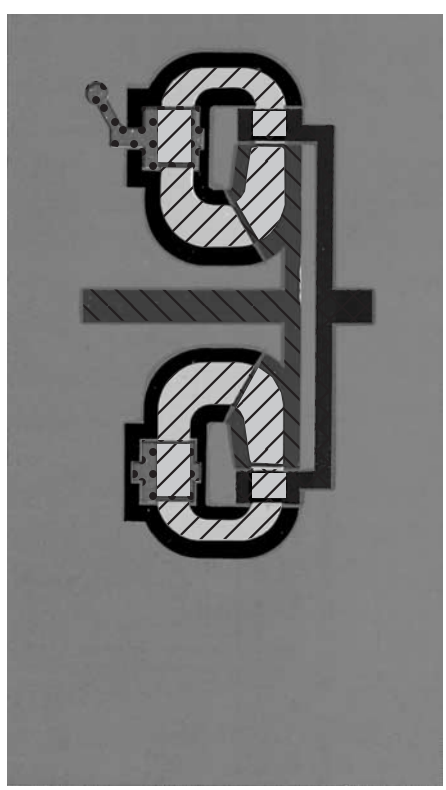
Up to size 487II (see Table P-13), these couplings are supplied as standard with an electrically driven oil pump. From size 487III upward, this pump is driven



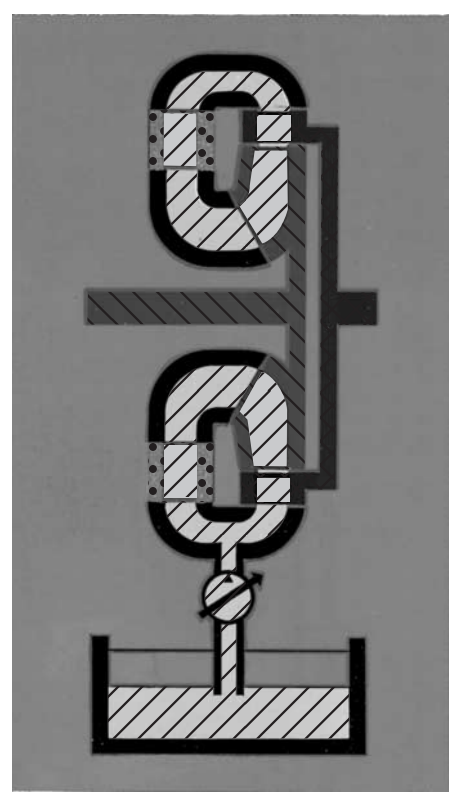
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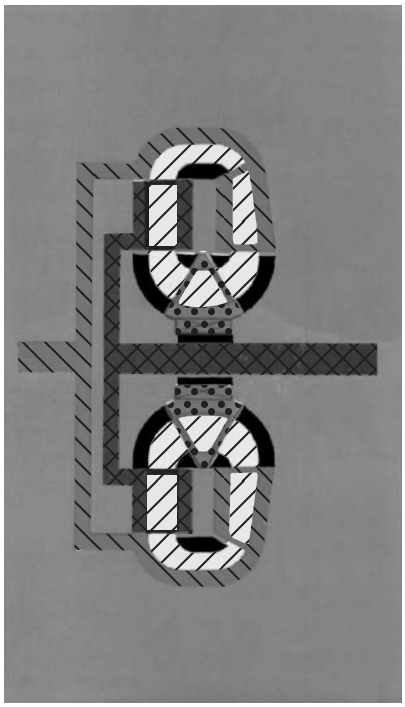
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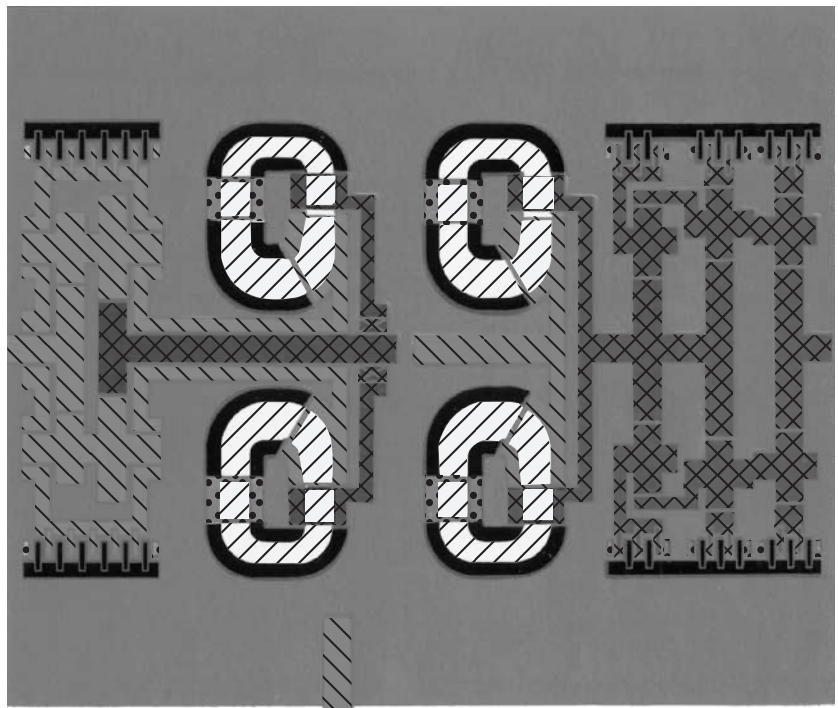
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FIG. P-104 The converter circuit. (Source: J. M. Voith GmbH.)

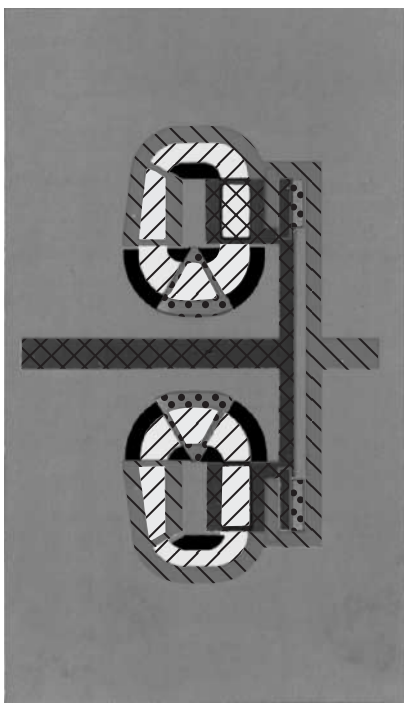
FIGS. P-105, P-106, P-107 Three methods of influencing the converter circuit. (Source: J. M. Voith GmbH.)



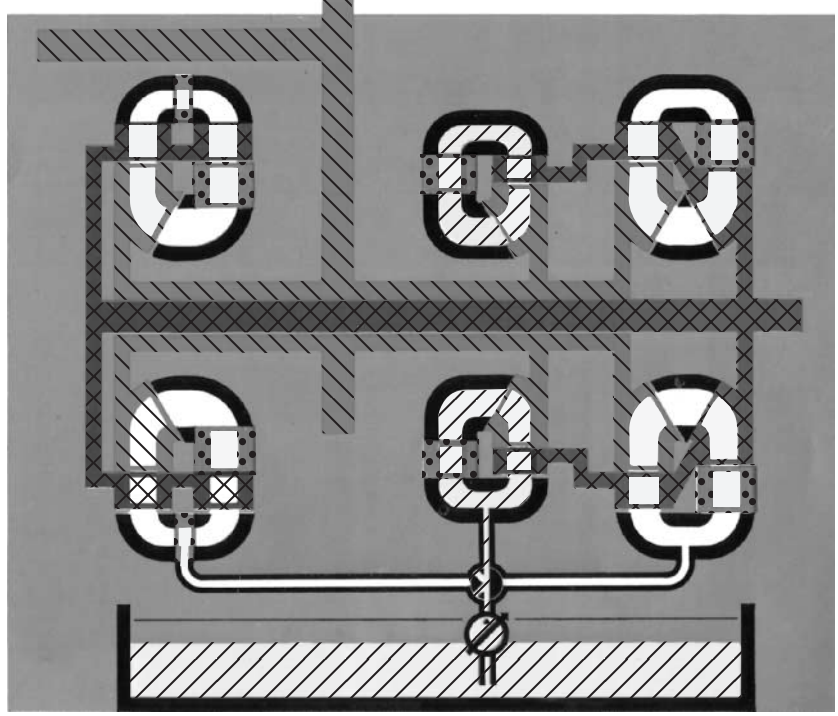
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FIGS. P-108, P-109, P-110, P-111 Influencing the converter circuit using mechanical elements. (Source: J. M. Voith GmbH.)

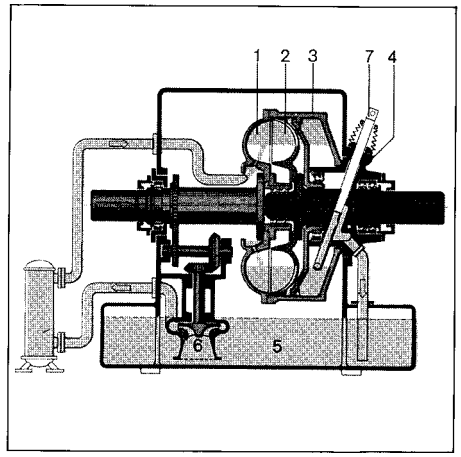


FIG. P-112 A simplified cross-section of a type SVNL hydrodynamic variable-speed coupling (self-supported, with the working pump gear driven from the primary shaft). 1, primary wheel; 2, secondary wheel; 3, shell; 4, scoop tube housing; 5, oil sump; 6, oil circulation pump; 7, scoop tube. (Source: J. M. Voith GmbH.)

TABLE P-13 Coupling Types in Service

Quantity	Type	Power of Each Pump	Pump Speed
10	562SVLIII	2700 kW	3495 rpm
3	R17K	3134 kW	4102 rpm
3	R15K550	4000 kW	6000 rpm
10	866SVNL	1460 kW	1445 rpm
2	R10K-API	802 kW	8950 rpm
4	R12K315-API	512 kW	5295 rpm
2	R12K315-API	683 kW	5026 rpm
3	R15K550-API	2980 kW	6200 rpm
2	464SVL-API	1193 kW	3585 rpm
3	464SVL-API	1388 kW	3564 rpm
1	R12K315-API	683 kW	5026 rpm

directly from the primary shaft by means of a mechanical gear. Deviations from this standard arrangement are sometimes necessary and can be provided in order to meet particular plant conditions.

Connections to the motor and to the driven machine are affected by flexible connecting couplings that allow for angular displacement. Figure P-113 illustrates transmission capacities in relation to speed.

Type SVL. Type SVL variable-speed couplings are suitable for driving fast-running machines of high power, such as boiler feed pumps, centrifugal compressors, and pipeline pumps (see Figs. P-114 and P-115).

To meet such requirements, the design features of this type of coupling include a strong cast housing designed to minimize vibration, plain bearings, use of steel castings for the coupling wheels, controlled circulation of the working fluid, and a lubrication system with a separate lubricating oil pump.

The working circuit includes the following major components: a scoop tube, scoop tube housing, heat exchanger, control valve, and return flow to the coupling. It follows that circulation is effected entirely by utilizing the dynamic pressure

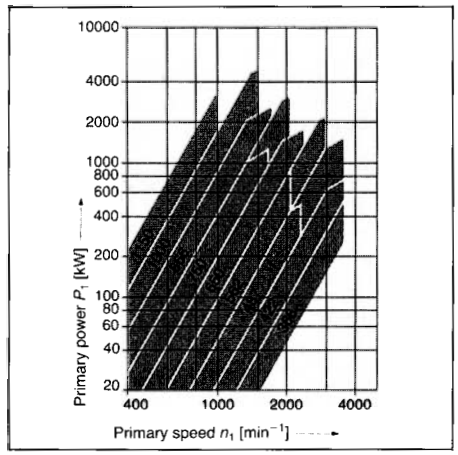


FIG. P-113 Power diagram for type SVNL variable-speed couplings. (Source: J. M. Voith GmbH.)

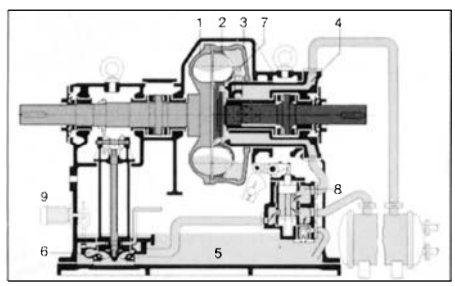


FIG. P-114 Cross-section of a type SVL hydrodynamic variable-speed coupling. 1, primary wheel; 2, secondary wheel; 3, shell; 4, coupling housing; 5, oil sump; 6, oil circulation pump; 7, scoop tube; 8, circulation control valve. (Source: J. M. Voith GmbH.)

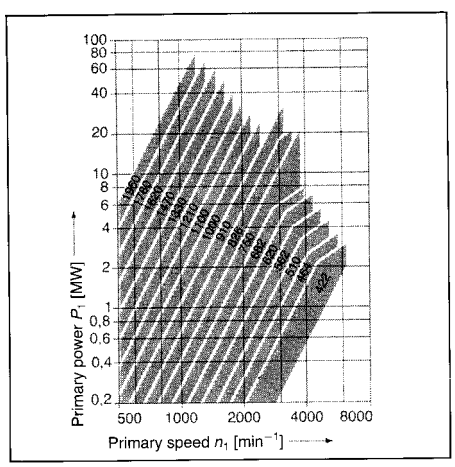


FIG. P-115 Power diagram for type SVL variable-speed couplings. (Source: J. M. Voith GmbH.)

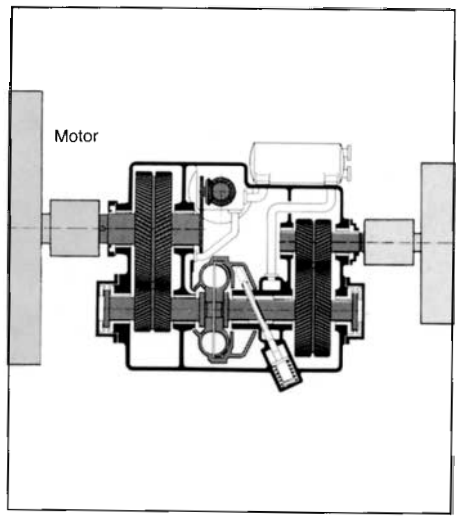


FIG. P-116 Cross-section of a type R KGS geared variable-speed coupling. (Source: J. M. Voith GmbH.)

generated at the mouth of the scoop tube. It is only changes in the quantity of oil in the coupling that are affected by the working pump and the control valve. This valve is actuated by a cam disk that is shaped to match the load characteristic. It provides optimal adjustment of the output of the oil circulation pump.

After matching the oil type, volume, and pressure, the lubricating oil supply to the motor, driven machine, and connecting couplings can be provided by the lubrication system of the variable-speed coupling.

Where particularly high output speeds are required, the variable-speed turbocoupling is equipped with two single-stage gears, one on the input side and one on the output side of the coupling. The result is the geared variable-speed coupling type R KGS (see Figs. P-116 and P-117).

Type RK. Geared variable-speed couplings combine a mechanical gear with a hydrodynamic variable-speed coupling in a common enclosure whose bottom half serves as an oil sump.

Starting from the basic concept R1.K (Figs. P-118 and P-119), comprising a speed increasing gear on the input side of a variable-speed coupling, various types have been developed to match such applications as infinitely variable-speed control of fast-running, high-powered driven machines such as boiler feed pumps or, with a speed-reducing gear on the output side, for the speed control of low-speed machines such as coal pulverizers, induced draught fans, and crude-oil pumps.

Geared variable-speed couplings are constructed for powers of up to 60,000 kW and more.

Case study 2: Pipeline pump stations and loading stations in Saudi Arabia. In the pumping stations 1 to 8 (Fig. P-120), a total of 52 geared variable-speed couplings are in service—often under extreme environmental and operating conditions. See also Figs. P-121 and P-122.

The advantages offered by the use of geared variable-speed couplings are:

- Energy savings compared with the throttling of centrifugal pumps when operating under partial load
- Design appropriate to the relevant explosion-proof regulations

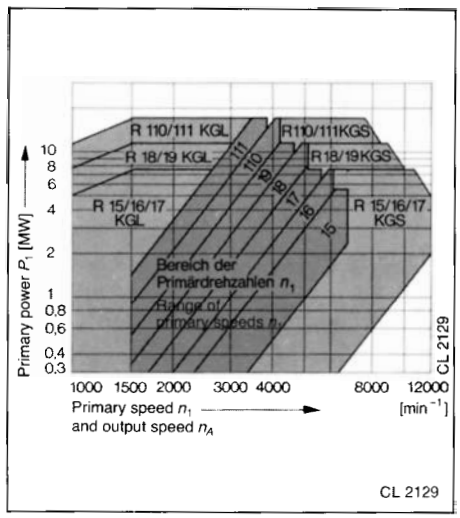


FIG. P-117 Power diagram for types R KGS and R KGL geared variable-speed couplings. (Source: J. M. Voith GmbH.)

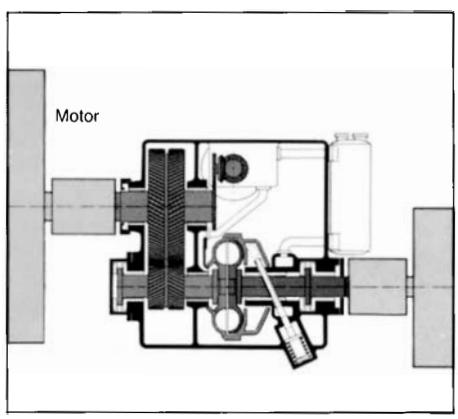


FIG. P-118 Cross-section of a type R1.K geared variable-speed coupling with speed increasing gear on the input side of a variable-speed turbocoupling. (Source: J. M. Voith GmbH.)

- Increased service life of the driven machine and other plant elements as a result of infinitely variable-speed control resulting in reduced wear
- Motor starting under no load followed by controlled load application to requirements
- Gentle acceleration of the driven machine by adjustment of the coupling filling
- Vibration and shock damping
- Wear-free power transmission by way of an operating fluid
- Disconnection of power transmission without stopping the prime mover
- Lubricating oil supplied to the motor and driven machine from the oil system of the geared variable-speed coupling

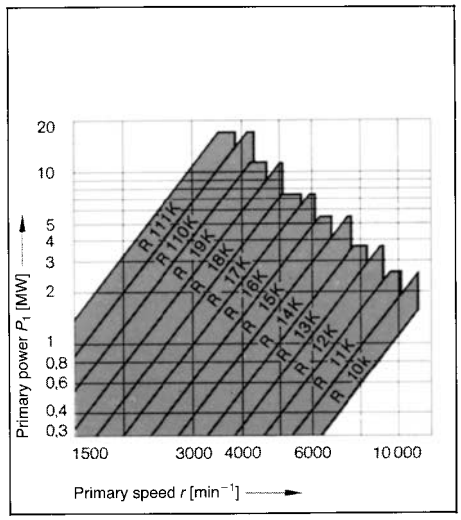


FIG. P-119 Power diagram for type R1.K geared variable-speed couplings. (Source: J. M. Voith GmbH.)

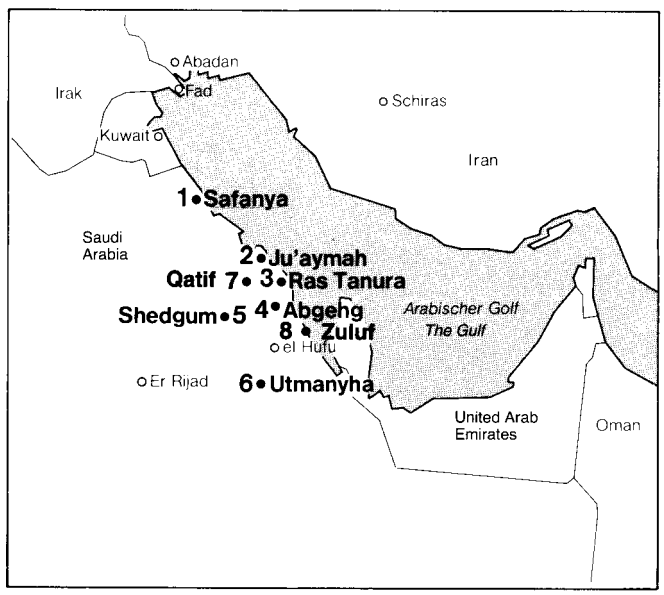


FIG. P-120 Pumps station sites in Saudi Arabia using geared variable-speed couplings. (Source: J. M. Voith GmbH.)

- Compact design with the gear units and the variable-speed coupling being contained in a single housing
- Low space requirement
- Economy in operation due to energy savings and also as a result of the motor and driven machine being subjected to less stress

The following features are of particular interest for the drive of crude-oil and liquid-gas pumps:

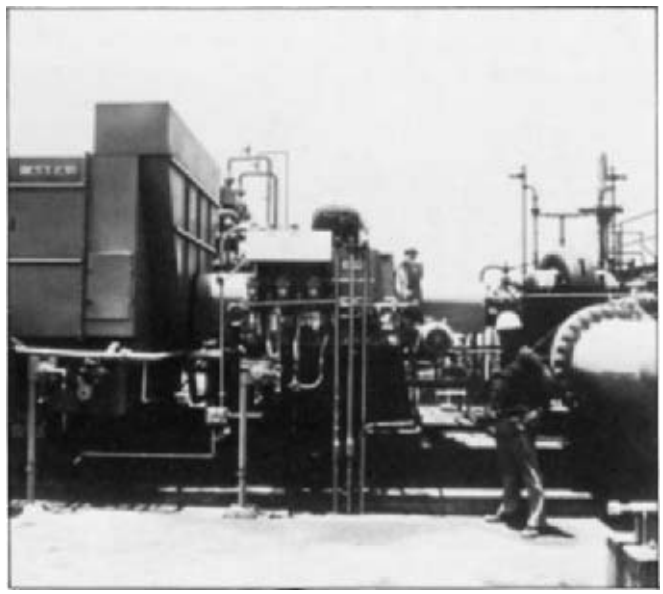


FIG. P-121 A bulk oil cargo terminal in the desert with a Voith type R17KGL geared variable-speed coupling. P_a = 7,000 kW, n_a = 1372 rpm. (Source: J. M. Voith GmbH.)



FIG. P-122 A bulk cargo oil loading station in Juaymah with a Voith R19KGL geared variable-speed coupling designed for the following data: P_a = 11,000 kW, n_a = 1800 rpm. In this station there are 11 units of the same size that are connected in series. They serve to pump crude oil to an offshore oil terminal. (Source: J. M. Voith GmbH.)

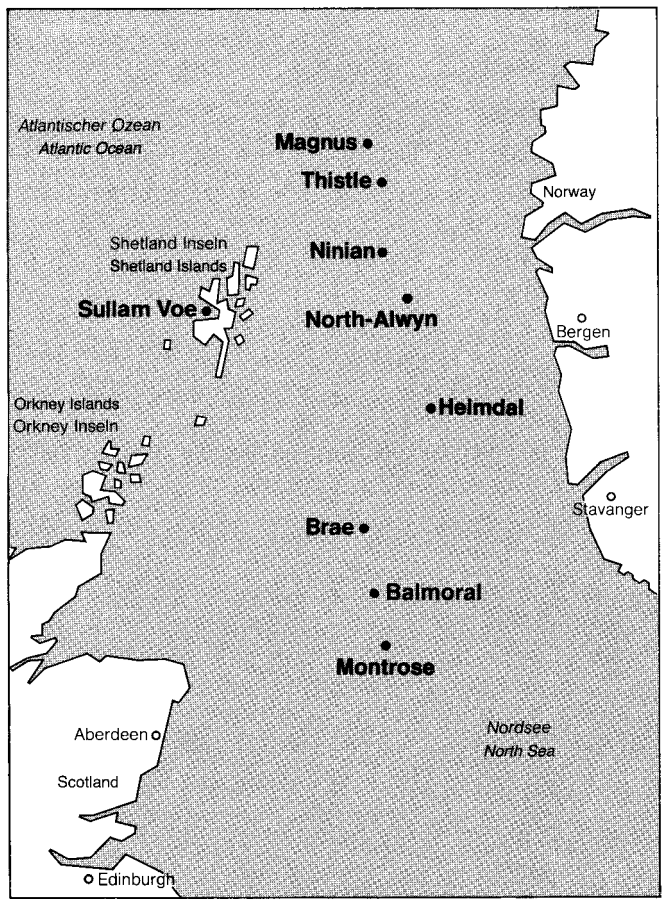


FIG. P-123 Some offshore platforms in the North Sea that use variable-speed turbocouplings. (Source: J. M. Voith GmbH.)

1. The possibility of adjusting the pump delivery by speed control, using a relatively simple control loop.
2. The possibility of keeping a motor running continuously and stopping the pump by emptying the coupling.

As is known, in bulk cargo oil terminals, the pump delivery must be kept constant irrespective of any variation in suction or delivery heads, and it is also necessary to keep the motors running during brief interruptions in delivery (such as on changing from one tanker to another) in order to avoid high peak starting currents. This is possible simply by draining the coupling. The secondary wheel is stopped by means of a brake.

Case study 3: Offshore pumping stations in the North Sea. A total of 16 variable-speed turbocouplings and geared variable-speed turbocouplings from this OEM are installed on a number of offshore platforms in the North Sea (Fig. P-123). Here the crude oil is pumped from a central platform through pipelines to the oil tanks on the Shetland Islands.

Speed control is necessary in order to maintain the flow against pipe friction that varies in relation to the quantity of oil delivered. Viscosity variations can also be compensated for by varying the pump speed.

The coupling types in service are depicted in Table P-14.

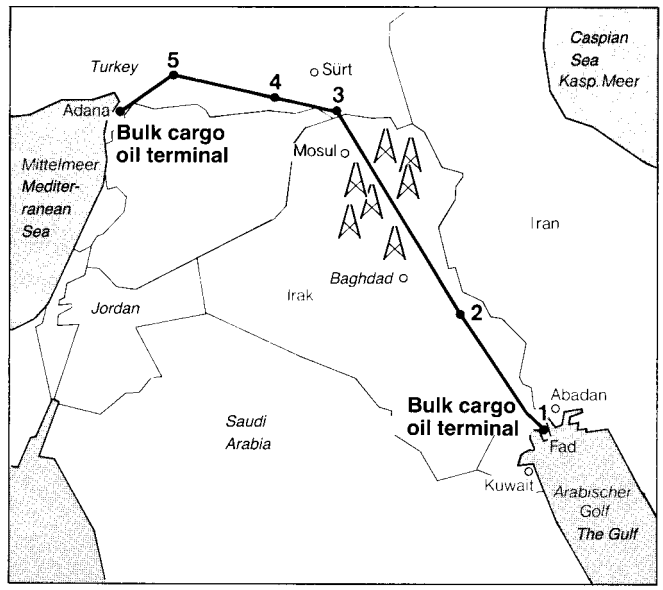


FIG. P-124 The course of the Iraq–Turkey pipeline. (Source: J. M. Voith GmbH.)

TABLE P-14 Information Source's New Installations Abroad

Application	Qty.	Type	Power	Speed
Oil pipeline in Russia	12	487 SVTL	1169 kW	2879 rpm
Water injection pumps in Holland	3	487 SVTL	820 kW	2908 rpm
Water injection pumps in Saudi Arabia	3	R19K600	6893 kW	4311 rpm
Pipeline Naftovod	6	R17K	2500 kW	4500 rpm
in Yugoslavia	4	562 SVNLG	1524 kW	2915 rpm
	4	487 SVNLG	800 kW	2917 rpm

Case study 4: Iraq–Turkey pipeline. To overcome pipe friction and changes in altitude along the path of a pipeline, it is generally necessary to install pressure-boosting stations.

In the case, for example, of the Iraq–Turkey pipeline with a length of around 1000 km, there are five pumping stations that are each equipped with four variable-speed couplings (Voith type 620SVLIII), each transmitting a power of 3400 kW at a speed of 2885 rpm. (See Figs. P-124 and P-125.)

At the bulk cargo oil terminal No. 1 in Fad, an additional two variable-speed couplings type 1,000SVNLIIV are in service, each transmitting 3000 kW at a speed of 1455 rpm.

General cases of the installations are listed in Tables P-14 and P-15 and shown in Figs. P-126 through P-128.

Case study 5: Energy savings example—boiler feed pumps. Energy saving on boiler feed pumps. Speed adaptation of boiler feed pumps by replacing the gear stages in the geared variable-speed turbocoupling. (See Figs. P-129 and P-130.)



FIG. P-125 A Turkish pumping station on the Iraq–Turkey pipeline with a Voith type 620SVL variable-speed turbocoupling with an oil/air oil cooler. (Source: J. M. Voith GmbH.)

Design

Motor power required: 8040 kW

Pump power: 7600 kW

Pump speed: 6500 rpm

This design was made in the planning stage considering most various construction variables with regard to piping, etc.

Maximum operating point

Motor power: 6160 kW

Pump power: 5300 kW

Pump speed: 5700 rpm

After some time it turned out that this operating point is never exceeded.

Optimized condition

Motor power: 5460 kW

Pump power: 5300 kW

Pump speed: 5700 rpm

Difference of motor powers: 6160 kW – 5460 kW = 700 kW

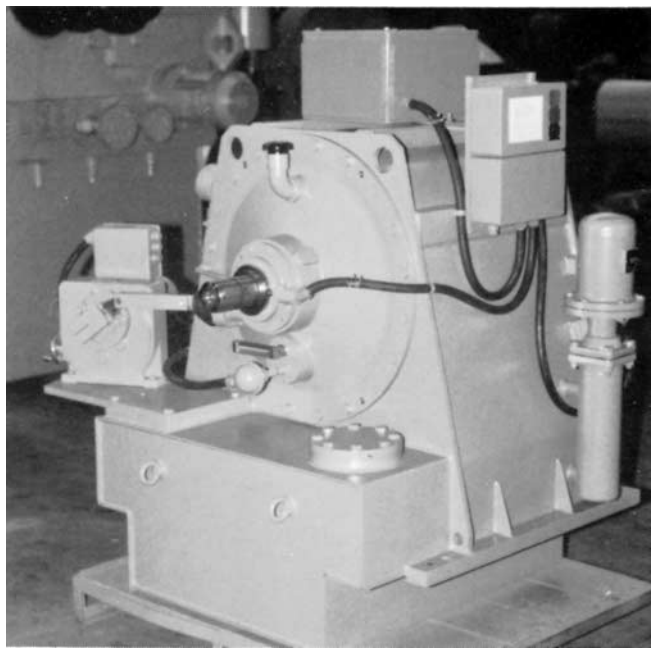


FIG. P-126 A Voith type 487SVTL variable-speed turbocoupling for a pipeline in Russia. (Source: J. M. Voith GmbH.)



FIG. P-127 A model of the water injection station at Qatif in Saudi Arabia. (Source: J. M. Voith GmbH.)



FIG. P-128 2 × R16K with gear units E45 arranged on the output side with air-oil cooler for driving a crude oil loading pump in Ras Tanuara, Saudi Arabia. Power consumption of pump 4000 kW, pump speed 1200 rpm. (Source: J. M. Voith GmbH.)

TABLE P-15 Information Source's Installations in West Germany

Refinery	Coupling Type	Rated Power	Speed
Vohburg	1 × 274SVN	30 kW	2980 rpm
Vohburg	2 × 316SVN	40 kW	2980 rpm
Vohburg	4 × 366SVN	15 kW	1450 rpm
Vohburg	2 × 422SVN	45 kW	1450 rpm
Vohburg	6 × 464SVL	1016 kW	4180 rpm
Vohburg	2 × 487SVNL	376 kW	2890 rpm
Vohburg	2 × 650 SVN	50 kW	740 rpm
Godorf	3 × 510SVL	540 kW	2980 rpm
Speyer	4 × 422SVN	258 kW	2980 rpm
Speyer	3 × 366SVN	120 kW	2980 rpm
Uerdingen	2 × 487SVN	55 kW	1450 rpm
Uerdingen	2 × 422SVNRI	37 kW	1470 rpm
Bucholtwelmen	3 × 422SVNL	255 kW	2980 rpm

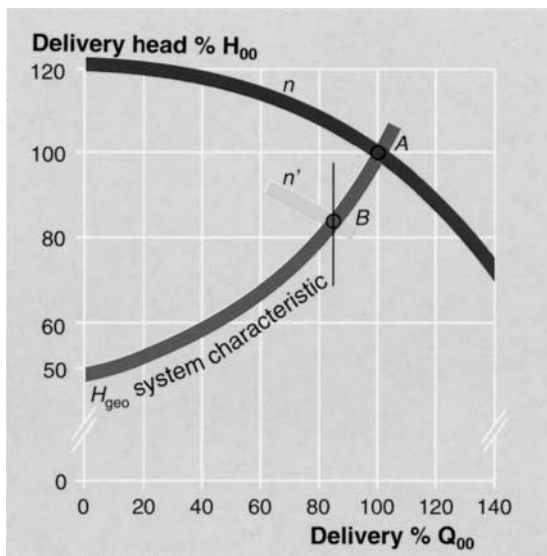


FIG. P-129 Q-H diagram with system characteristic and pump characteristic—design point A and speed-regulated condition B. (Source: J. M. Voith GmbH.)

This represents the continuous energy saving during operation. With an electricity price of 0.12 DM/kW hour and just 6000 operating hours per year, there is an annual saving of 500,000 DM.

Case study 6: Speed regulation using hydrodynamic variable-speed couplings in place of throttle regulation. This case is a summary and discussion of the various economic factors that influence the choice between speed and throttle regulation. A power station boiler feed pump is used as an example but the argumentation equally applies to all other drives with a parabolic or near-parabolic relationship between torque and speed, e.g., all types of circulating pumps and fan drives.

Principle of throttle and speed regulation of boiler feed pumps in power stations. As Fig. P-131 shows, the operating point is the point where the boiler resistance curve and the characteristic pump curve (n = const.) intersect. To operate at a different point, either the boiler resistance curve or the pump curve must be

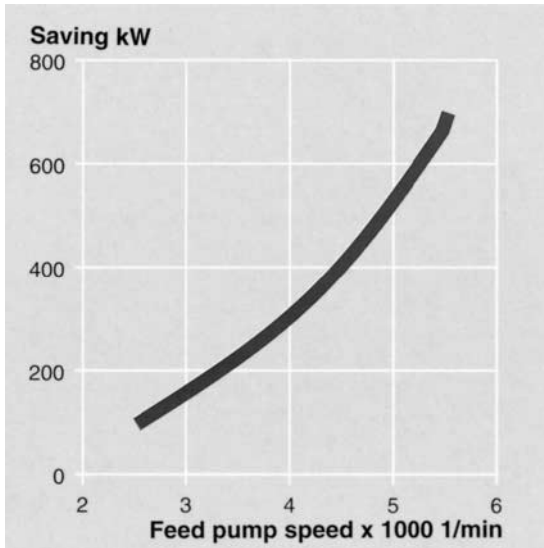


FIG. P-130 Energy-saving curve with optimized ratio. Example: boiler feed pump drive. (Source: J. M. Voith GmbH.)

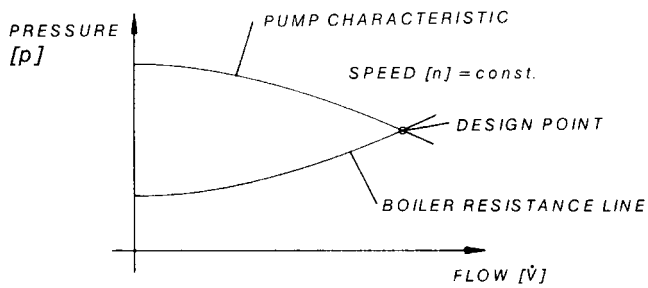


FIG. P-131 Pump characteristics and boiler resistance curve. (Source: J. M. Voith GmbH.)

changed. This has led to the development of two regulating methods that are now in general use:

a. By installing a throttle valve, the boiler resistance curve is brought up to the required higher pressure level (throttle regulation).

b. By varying the pump speed, the pump curve is changed (speed regulation).

Design of the boiler feed pump. The maximum continuous load of boiler must be the basis for determination of the design point of the feed pump (see Fig. P-132).

1 = Boiler resistance curve in constant pressure operation
2 = Boiler resistance curve in varying pressure operation
3 = Pump curve of the half-load pump
4 = Pump curve of the full-load pump
5 = Maximum continuous rating of the boiler
6 = Design point
Pk = Maximum permissible boiler pressure

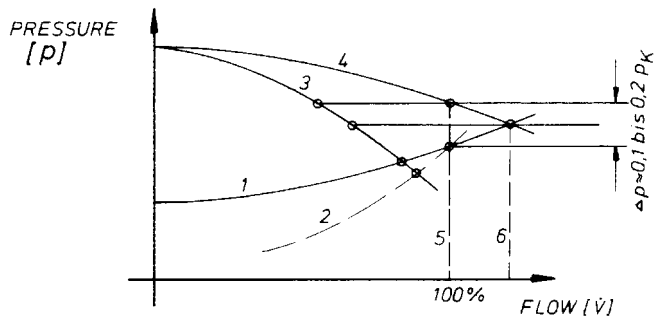


FIG. P-132 Curves required for boiler feed pump design. (Source: J. M. Voith GmbH.)

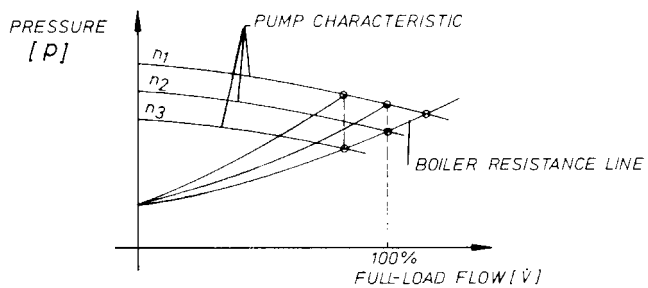


FIG. P-133 Optimizing pump performance. (Source: J. M. Voith GmbH.)

The pressure head of the feed pump must be so sized that the geodesic head can be overcome and the required pressure increase, flow rate, and pressure losses plus a 10 percent extra (related to the maximum boiler operating pressure) can be attained.

For completeness, Fig. P-132 also has the curves for constant and varying pressure operation. The fact that the position of these curves differs has no real effect on the subject under discussion. Furthermore this introductory section only deals with the basic outline of the considerations that affect the choice of a feed pump.

Comparison between speed regulation and throttle regulation

OPTIMUM USE OF PUMP. Figure P-133 shows that with throttle regulation, the throttle valve has to raise the boiler resistance curve even during full-load operation. Each flow rate reduction required can only be achieved by artificially further raising the boiler resistance curve. The delivery throttling therefore decreases the unit's efficiency since the power throttled off cannot be recovered.

With speed regulation of the pump only a relatively small speed change is required to cover the whole boiler load, and the pump efficiency does not change very much. Furthermore when the delivery rate drops the speed and pump loading also drops, this means extended pump life due to reduced wear.

INCREASED PLANT LIFE. It is a general rule for all pump units (not only in power stations) that speed regulation has a beneficial effect on performance and service life, because the operating conditions in the pump are more favorable. If high-pressure pumps in particular are run for a long period with the flow heavily throttled back, overheating can occur leading to wear on the races, shaft bushes, seals, etc. The other units around the pump are also loaded in that energy is put

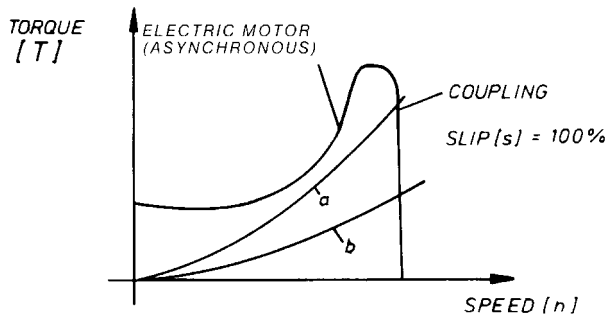


FIG. P-134 Torque versus speed for an asynchronous motor. (Source: J. M. Voith GmbH.)

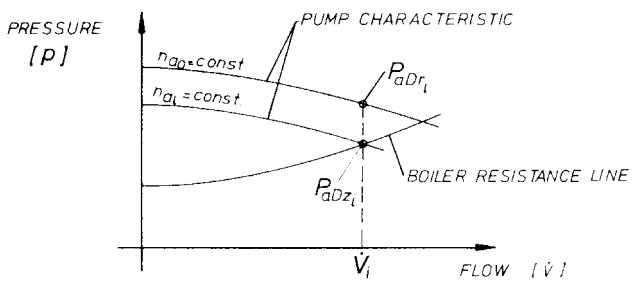


FIG. P-135 Flow versus pressure pump characteristic. (Source: J. M. Voith GmbH.)

into the flow and then destroyed by the throttle valve or transformed into heat; of course the valve is itself very heavily stressed.

ENERGY SAVINGS. As already mentioned, speed regulation saves energy that is dependent on the load schedule and the pump and plant curves. A detailed, quantitative analysis of these savings follows under "Efficient Energy Utilization with Speed Regulation."

EASIER MOTOR STARTUP. When an electric motor starts up, problems occur that come under the heading breakaway torque, startup current, and motor design in accordance with the startup conditions. The hydrodynamic coupling represents a proven means of overcoming these problems.

Even the most unfavorable case, namely a blocked driven machine, would only subject the motor to a load that rises to the square of the motor speed (see Fig. P-134). This assumes the coupling is full (curve *a* in Fig. P-134) whereas in practice the coupling filling is set to minimum during startup (curve *b*). In the case of extreme requirements, the favorable behavior of the coupling is backed up by an integrated hydraulic device, usually called a rapid start device. This means that the motor can have a large acceleration capability at its disposal and really only needs to overcome the inertias of the motor armature and coupling wheels. Controlled loading by automatic filling of the coupling does not start until the most favorable rated torque on the motor curve has been reached.

Since the hydrodynamic coupling is the connecting member between the motor and the driven machine, it influences not only the motor-side startup conditions but also those of the driven machine. By connecting the coupling's scoop tube to suitable control instruments or closed loops, it is possible to carry out complicated operations, such as torque limitation, acceleration limitation, and startup time

reduction. In the case of boiler feed pumps the ability of the standby unit to start up very rapidly plays a major role in protecting the whole plant. This is one of the reasons why the hydrodynamic variable-speed coupling has become such an important element in power stations, since this condition is met simply by connecting up the scoop tubes of two couplings in parallel and adapting the hydraulic or electric control circuit accordingly.

The dynamic behavior of coupling together with the units they are connected to is by now very well known, so the designer can be supplied with suitable data.

SEPARATION OF DRIVING AND DRIVEN MACHINES. Separation of the driving and driven machines can mean vibration and shock separation or absorption; it can also include the possibility of power separation by draining the coupling. At all events the hydrodynamic variable-speed coupling can optimally satisfy both requirements that are of major importance as far as operating performance is concerned. Because of the present-day speed and performance demands made on machines, shock and vibration absorption is an important factor, not only when reciprocating with rotating machine elements work together. The drainability of the coupling is mostly used to protect the prime mover, i.e., to relieve the load on the motor, to avoid long-duration startup current peaks, and to reduce the load on the mains. It is often the case that large motor units are not shut down when the driven machine is to be stopped and that frequent motor startups, and thus mains loading, are avoided simply by draining and filling of the coupling. This means the plants are more economical because of increased protection of the electrical and mechanical drive elements even with very high startup frequency.

This subsection was a summary of the various factors that illustrate the advantages of speed regulation using a hydrodynamic variable-speed coupling in feed pump drives and that can be applied to these drives. Some of the factors mentioned affect economy of operation indirectly rather than directly, e.g., reduced breakdown rate, different design data for the whole plant, operating safety, and flexibility.

Efficient energy utilization with speed regulation. To calculate the energy savings achieved by speed regulation using a hydrodynamic variable-speed coupling, the unit resistance curve, the characteristic pump curve for various speeds, and the load schedule must be known. The unit and pump curves are specified parameters and thus known within certain tolerances. This is not the case with the load cycle, which depends on the interrelationship with other power stations, the power increase or requirements over a large number of years, the block size, etc. It is very difficult to make correct assumptions because the interdependent factors are highly complex and the developments have to be assessed for a long time ahead because of the normal amortization periods for large plants.

However the following calculation shows that speed regulation pays for itself very quickly simply due to the throttle energy saved. It is therefore the author's opinion that a decision in favor of speed regulation is not really as difficult as it first appears.

The power consumption of the pump at the individual operating points i corresponds to the following formula; the separate values for V_i and p_i out of the unit resistance curve and the pump curve can be deduced.

$$P_a = \dot{V}_i \cdot \frac{p_i}{\eta_i} \tag{P-5}$$

P_a = pump power consumption with throttle regulation
$\dot{V}_i$ = oil flow
P_i = pressure
η_i = pump efficiency

The energy difference per pump unit is as follows:

$$\Delta E_{total} = \sum_{i=1}^{n} (P_{Dri} - P_{Dzi})\, i \cdot t_i \qquad \text{(P-6)}$$

P_{Dri} = power requirements with throttle regulation
P_{Dzi} = power requirements with speed regulation
t_i = startup frequency at flow V_i (h/year)

where the following applies:

$P_{Dri} = P_{aDri}$ (pump power consumption with throttle regulation; see Fig. P-135).

Taking into consideration the efficiency and slip of the fluid coupling, the resultant power requirements with speed regulation are:

$$P_{Dzi} = P_{aDzi} \cdot \frac{n_{ao}}{n_{ai}} \cdot \frac{100}{100 - s_{min}} (1 + a) \qquad \text{(P-7)}$$

P_{aDzi} = pump power consumption with speed regulation (see Fig. P-136)
n_{ao} = (maximum) pump speed with throttle regulation
n_{ai} = pump speed with speed regulation
s_{min} = minimum slip of fluid coupling in percent
a = symbol for mechanical efficiency of coupling (0.02 – 0.03)

Since Eq. (P-7) is an approximate formula for what is in reality a difficult physical process, Eq. (P-7) should only be used as a rough calculation. Values a and s_{min} in Eq. (P-7) are average values reflecting the fluid coupling's efficiency, which is in fact dependent on the coupling size and type and the load. However the formulae are accurate enough for our purposes and, being simplified, make it easier to understand the basic interrelationships.

NUMERICAL EXAMPLE. The previous description of the economical factors is here further clarified using a numerical example. Figure P-136 illustrates the following calculations.

The example is based on a real medium size power station; some values that are valid for this particular power station have been modified to make them of more general application. The author believes that the data were selected to the advantage of throttle regulation; this is dealt with in more detail in the following section where the design values are discussed. This example is mainly meant to be an example that can be quickly amended by introducing further data.

The throttle curve in Fig. P-136 is, because of its flat rise, more or less an ideal throttle regulation curve.

Such a curve cannot always be achieved because in practice too flat throttle curves mean instable pump behavior. Cases where the throttle curve does rise more acutely mean higher energy losses for throttle-regulated plants.

The pump is normally designed so that maximum efficiency is achieved at the point where the pump most often operates; this we have called normal point B_N. This applies both to throttle and speed regulation. However, with speed regulation, the design efficiency can be maintained with very little deviation over a wide regulating range whereas with throttle regulation the efficiency drops sharply as the throttle regulation increases.

Three operating points are given in Table P-16.

The following are assumed: feed water temperature of 165°C, specific gravity of 900 kg/m^3, and motor speed of 1480 1/min.

The appropriate coupling for these values is the hydrodynamic variable-speed geared coupling type R 16 K-550. For clarity arrangement sketches have been added

TABLE P-16 Operating Points

Operating Point	m (kg/s)	p (bar)	n_a (1/min)	P_a (kW)	t (h/year)
Max. point B_M	172.5	209	5980	5348	50
Norm. point B_N	155	199.7	5870	4417	5000
Min. point B_S	122	182.2	5500	3236	2800

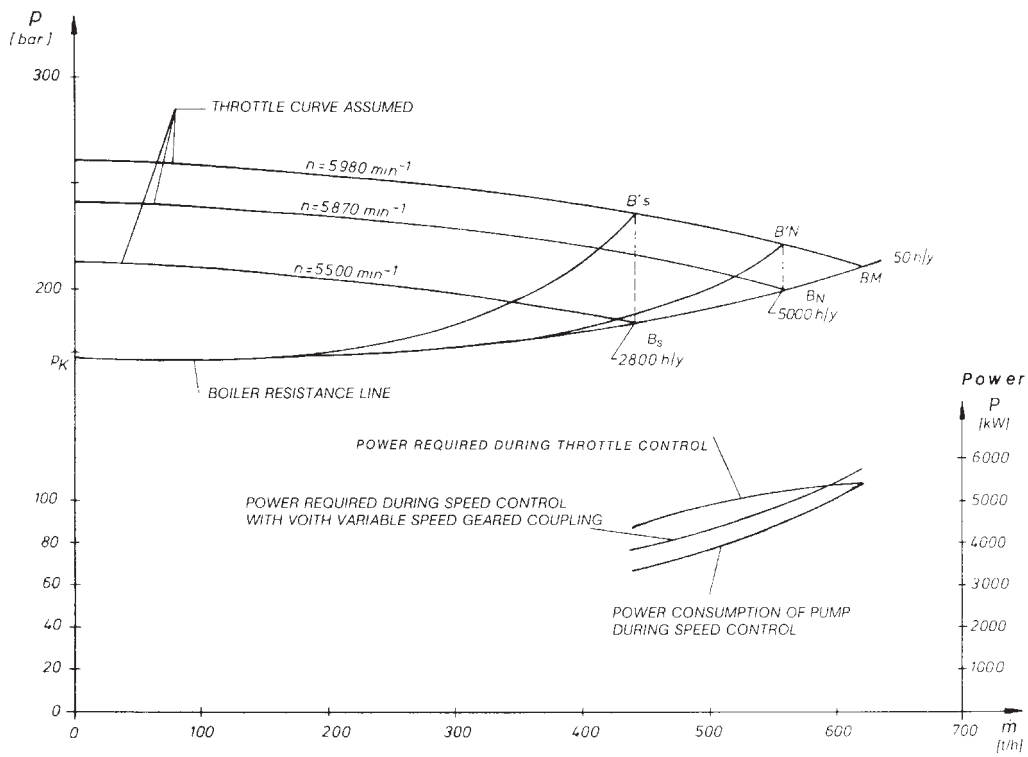


FIG. P-136 Comparison of throttle control and geared coupling control. (Source: J. M. Voith GmbH.)

(Figs. P-137 and P-138) showing the motor, feed pumps, and the coupling that combines a mechanical gear and a fluid coupling.

The theoretical energy saving is opposed by the efficiency losses of the coupling, detailed in Table P-17 as slip, and the resultant hydraulic and mechanical losses (bearing friction, etc). The power losses were calculated by means of a table computer program (see Table P-17).

We have assumed the pump efficiency for throttle regulation operation at the corresponding points to be as follows:

Maximum performance $B_M\eta = 0.76$
Normal performance $B_N\eta = 0.74$
Minimum performance $B_S\eta = 0.66$

$$P = \frac{V \cdot p}{\eta} = \frac{m \cdot p}{\rho \cdot \eta}$$

TABLE P-17

Operation Point	s (%)	P_{vs} (kW)	P_{vm} (kW)
Max. point B_M	2.7	148	149
Norm. point B_N	4.5	208	155
Min. point B_S	10.5	382	172

P_{vs} = hydraulic losses; P_{vm} = mechanical losses in the coupling; s = slip.

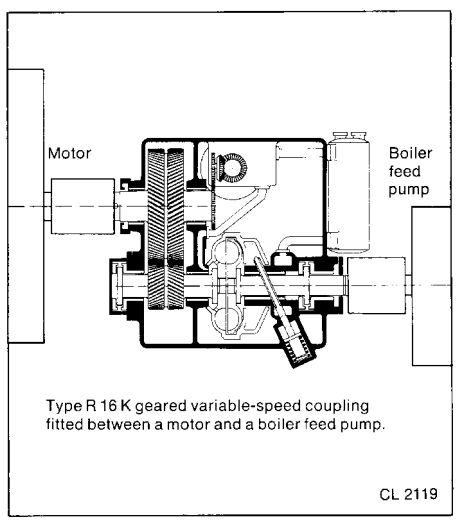


FIG. P-137 Arrangement sketch: geared variable-speed coupling. (Source: J. M. Voith GmbH.)

Table P-18 shows the result for the three load points: column one gives the performance with throttle regulation using the above formula, column two the performance with speed regulation taking account of the losses in the fluid coupling, and column three the difference between the two.

It is immediately obvious from these results that throttle regulation is only favorable for maximum power operation. However, this operating point is only used under certain conditions and, even then, very rarely; this is why the second point is called the normal point. When one considers the normal point and the minimum power operating point, the advantage of speed regulation using a hydrodynamic variable-speed coupling is unmistakable.

To be in a position to assess the difference between the two types of regulation from the financial point of view, we must first establish an annual load schedule. The schedule assumes that the power station block runs for 7850 hours per year with 5000 hours at normal power. This assumption again favors throttle regulation since power station blocks of this rating are used more for peak or average loads and not, as assumed here, for base loads. The results in MWh are given in Table P-19.

The 3757.15 MWh savings achieved by speed regulation using a hydrodynamic variable-speed geared coupling means that the fluid coupling pays for itself in less than a year, assuming a price per kWh of 0.03 DM.

The energy price is highly dependent on fuel costs, plant size, commercial valuations, etc., but the assumed value should represent a reasonable mean value.

TABLE P-18 Results for Three Load Points

Operating Point	Throttle Regulation (kW)	Speed Regulation (kW)	Throttle Coupling (kW)
Max. point B_M	5348	5645	–297
Norm. point B_N	5120	4780	340
Min. point B_S	4530	3790	740

TABLE P-19

Annual Op. Hours	Throttle Regulation (MWh)	Speed Regulation (MWh)	Throttle Coupling (MWh)
B_M = 50	267.40	282.25	–14.85
B_N = 5,000	25,600.00	23,900.00	1,700.00
B_S = 2,800	12,684.00	10,612.00	2,072.00
Total 7,850	38,551.40	34,794.25	3,757.15

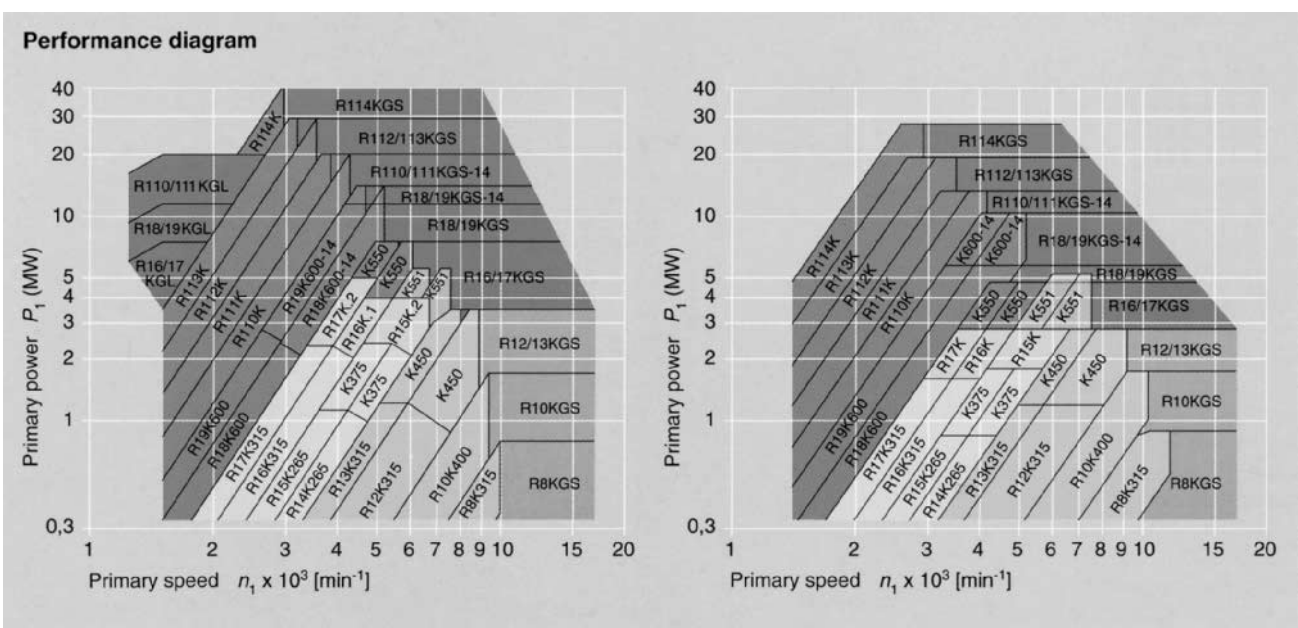


FIG. P-138 Performance diagram: geared variable-speed coupling. (Source: J. M. Voith GmbH.)

The material in this case study resulted in a recommendation in favor of the speed regulation of power station feed water pumps by hydrodynamic variable-speed couplings. The extra unit pays for itself within approximately one year. A decision in favor of speed regulation is indicated. Factors such as power station blocks designed for based loads (i.e., high-performance blocks) being used for peak and medium loads 5–10 years later support the decision.

TABLE P-20 Equipment and Design Characteristics

	Gear	Pump	Scoop Tube	Oil Flow
R8-13K	Single-stage, single helical	Gear pump	Hydraulically operated	Constant
R8-13KGS	Double-stage, double helical	Gear pump	Hydraulically operated	Constant
R . . . K	Single-stage, double helical	Pump combination	Direct mechanically operated	Variable
R . . . K551	Single-stage, single helical	Pump combination	Hydraulically operated	Variable
R . . . K265 R . . . K375	Single-stage, single helical	Gear pump	Hydraulically operated	Variable temperature controlled
R . . . K550/ 600/630	Single-stage, double helical	Pump combination	Hydraulically operated	Variable
R . . . KGS R . . . KGL	Double-stage, double helical	Pump combination	Hydraulically operated	Variable

Slip losses are small; they do not affect the efficiency of the total power station much. However, consideration was paid to the possibility of bringing the slip losses back to the working circuit, e.g., the power station circuit or the central heating circuit by means of a heat exchanger. This means that speed regulation with the hydrodynamic coupling proved itself superior to throttle regulation, and that speed regulation has been improved.

Coupling selection

The design of a geared variable-speed coupling is mainly determined by the input power and input speed of the driven machine. The performance diagram shows the power transmitted by various coupling types and sizes, depending on the input and output speeds.

Table P-20 shows varying equipment and design characteristics of the individual types of geared variable-speed couplings. This information source has couplings for individual applications. Matching coupling combinations are created by arranging the gear stage and the coupling in relation to the required power, motor speed, and speed of the driven machine. Coupling types are generally as follows (see Fig. P-139):

1. A step-up gear is located in front of the coupling. This allows adaptation of high-speed machines to the speeds of 2- or 4-pole squirrel cage motors (Type R . . . K).
2. The coupling is driven direct and adapted to the speed required by the driven machine by a step-up gear situated after the coupling. For high powers at very high speeds (Type R . . . GS).
3. A step-up gear is located both at the input and the output side. This is suitable for particularly high output speeds, e.g., with high-speed boiler feed pumps (Type R . . . KGS).
4. A step-up gear at the input side and a reduction gear at the output side make this type into a fast reacting and compact coupling for fast response times (Type R . . . KGL).

Next Page